

Platform economy and the (partner) platform ecosystem

How do platform businesses differ from pipeline businesses?

What's the value of ecosystems in the platform economy?

December 2024 & January/February/March 2025

Disclaimer: The content in this document is largely based on the book "Platform Revolution". One of the authors turned the work into a C-level consultancy firm that provides a guide to platform business models about which Jim Hagemann-Snabe (Chairman of Maersk and Siemens & former CEO of SAP) says: "An inspiring guide for business leaders to transform existing businesses to platform businesses." [The Platform Institute | Guide to Platform Business Models](#) (video, 8 minutes). This is a very condensed/simplified view of the platform economy & a platform ecosystem.

Summary



To start with, it's worth making a [distinction](#) between platform from a technology perspective and from a business model perspective:

- A platform business is a **business model** (not a technology infrastructure) that focuses on helping to facilitate interactions across a large number of participants. The role of the platform business is to provide a governance structure and a set of standards and protocols that facilitate interactions at scale so that [network effects](#) can be unleashed. [Deloitte](#)
- A platform company is a type of business that provides a **technical foundation** and/or set of applications and tools which can then be used by other companies, either in the day-to-day running of their business, or to create their own products and solutions [on top of the platform](#) company's core offering. [Infobip](#)

Unlike non-asset platform companies (Uber, AirBnB), incumbent B2B companies seem to be more like Apple or Samsung who have built a [platform business on top of their pipeline business](#). A [platform business](#) is characterized by a triangular supply chain

(ecosystem of supply, demand, firm without owning all resources) while pipeline businesses have a linear supply chain owning all assets. In addition to that, [4 types of platforms](#) have been identified for B2B businesses, [REDACTED]

1. [Technology enablement platforms](#),
2. [Intelligent products and services](#),
3. [Data aggregation and collaboration platforms](#)
4. [Marketplaces](#)

Having multiple platforms is not uncommon. According to a panel discussion at the [World Economic Forum in China in 2018](#): *"You should never speak about the Microsoft or the Facebook platform or the [Google](#) platform. They have about 30 or 40 different platforms that sit underneath. Each of them with its own SBU (strategic business unit) style level project on what do we do in this particular submarket and with some connection between them in terms of my desire to be in particular areas. **So, we are revisiting the corporate strategy level at the platform level. Which means that we do things at the business level, and we see how they connect**".*

[REDACTED]

[REDACTED]

There are [3 major shifts](#) from a business standpoint in the [platform economy](#):

- From resource control to resource orchestration
- From internal optimization to external interaction
- From focus on customer value to focus on [ecosystem value](#)

which are supported by the [3 structural aspects](#) of the platform disruption:

1. De-linking assets from value
2. Re-intermediation
3. Market aggregation

De-linking assets from value might lead to less product sales and/or lower prices in some cases (sharing economy, [MRI example](#)). To substitute this loss in earnings, a company might decide to switch to a platform business model and provide [services](#) on top of the product (technology platform) and/or aggregate a dispersed market to benefit through re- intermediation. What makes platform businesses win over pipeline businesses is the so-called [network effect](#). Apparently, *"you can't scale network effects inside as easily as outside."*

[REDACTED]

[REDACTED]

[REDACTED]

Also, unlike [AirBnB](#) that added spare capacity (spare room) to a market that had more demand than supply (lack of hotel rooms during a design conference in San Francisco in 2007) and turned it into a business that aggregated the dispersed market of (initially) cheap, unused bedrooms through re-intermediation, B2B looks a little different. Partners need to be trained and onboarded to do business with an incumbent B2B company's products/solutions, no matter whether they join the [marketplace or not](#). The supply side needs active enablement. This is handled by [partner management](#). Because: *"In the B2B space, we do not see strong winner-takes-all characteristics yet," Dartmouth College professor Geoffrey Parker said at the [2023 MIT Platform Strategy Summit](#). "B2B tends to have a lot of variation, tremendous integration costs, and a complex environment that makes it harder to have the kind of scale that you see in the B2C world."*

However, there is something in the industrial space that might be considered "spare capacity": IoT data. *The network effect also applies to M2M (machine-to-machine) nodes. Assuming such a network effect will take place, one can assume it will create a huge value for the ecosystem ([Silent Intelligence](#)).*

What does that mean for digital marketing and digital experience?

Effective platforms are able to expand in size quickly and easily, thereby scaling the value that derives from network effects.

Everyone has heard of Instagram's rapid growth—over 100 million active users in less than two years—which led to its billion-dollar acquisition in April 2012 by Facebook. What is less well known is that Instagram achieved this rapid growth without employing a single traditional marketing manager. It happened because the company's platform was carefully designed to make viral growth organic and practically inevitable.

[REDACTED]

However, looking at fundamental change happening in the industrial space regarding processes ([from functional silos to data-driven processes](#)) and the customer research conducted [REDACTED] is a big challenge for customers, another approach to incorporate might be [Customer Education Marketing](#) which could be applied in both the presales phase ("educating the market") as well as in the post-sales phase when it comes to product adoption.



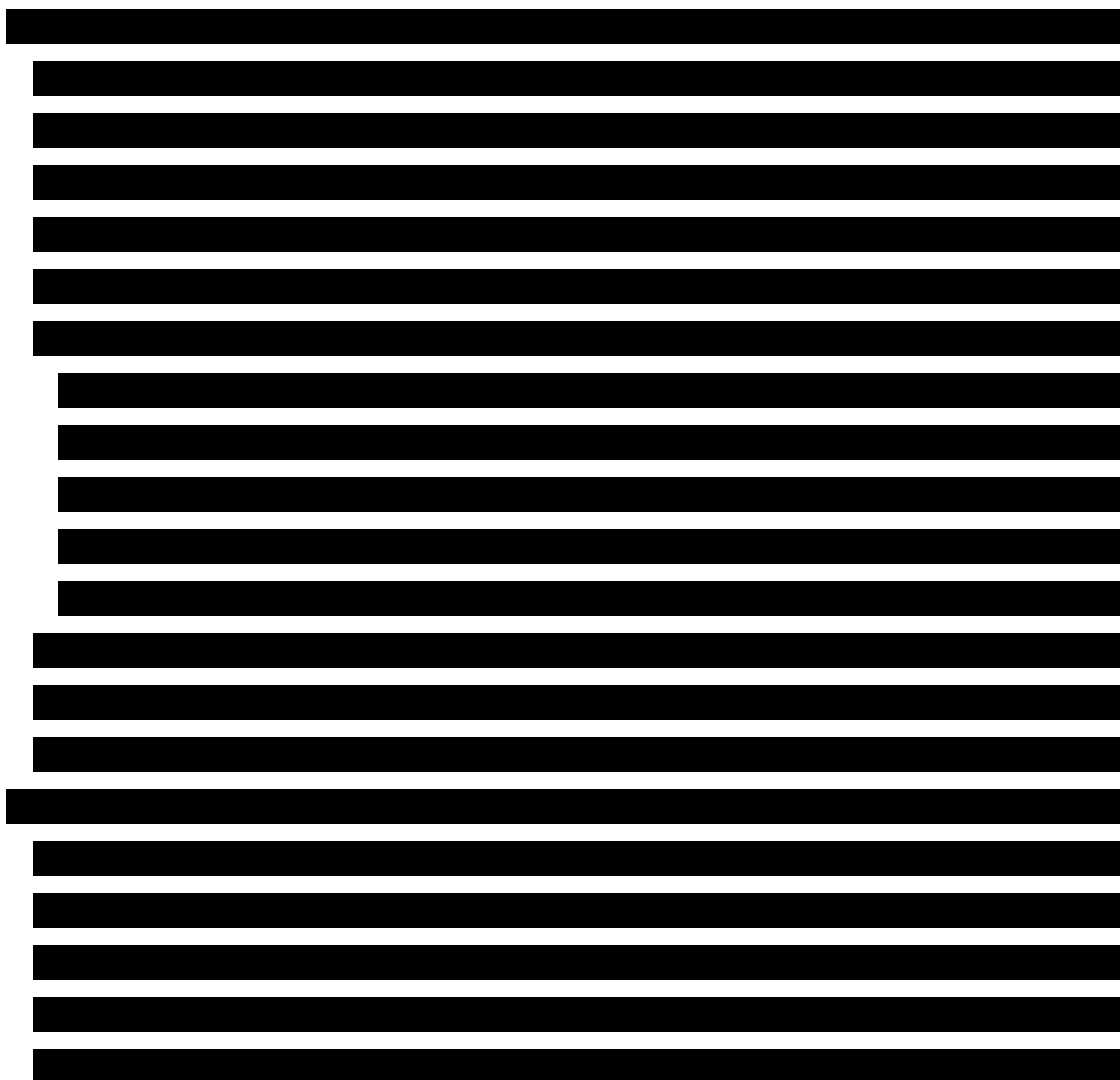
Questions

- Why would a partner point their customers to the marketplace (virality) where the partner faces competition from other partners (potentially in cheaper labor countries) instead of becoming another [re-intermediation instance](#) and offering a “search service” to their customers for products they do not offer? Or is this how the marketplace is supposed to work?
 - o Does the marketplace handle transactions (fulfillment) for partners?

[illegible]

Summary.....	1
Questions.....	5
Definitions.....	9
Marketplace platform.....	10
Technology platform.....	11
Development platform.....	12
What is a platform economy?.....	12
How economic value creation & processes changed with each industrial revolution..	13
The 3rd industrial revolution: Efficient pipelines eat inefficient pipelines.....	14
The 4th industrial revolution: Platforms eat efficient pipelines.....	15
Pipeline vs platform business model.....	16
How value is created in pipeline vs. platform businesses.....	17
Economic impact.....	18
Network effects.....	19
Monetization of network effects.....	21
Organizational impact.....	22
Dynamic capabilities.....	23
Is it one (platform) or the other (pipeline)?.....	26
Beyond the marketplace: The competitive advantage of technology platforms (speed of innovation).....	27
How does this work in B2B?.....	28
Four types of B2B platforms.....	30
Example: Energy market aggregation.....	31
Why is the industrial space lagging?.....	31
And what about hardware?.....	32
Example: GE Aviation.....	33
What other business platforms exist in the industrial space?.....	34
The Platform Manifesto.....	35

What is an ecosystem in the platform economy?.....	35
An ecosystem: Organizations of organizations.....	36
What does this look like?.....	38
What's good for the platform is bad for the ecosystem.....	40
Communities in an ecosystem.....	41
Community management is the new human resources management.....	43
DX is the new UX.....	44
Example: SAP community incentives & roadmap for developers.....	45



	11
Marketing & platform ecosystems.....	54
Marketing a platform.....	54
What is the product?.....	54
Example: Spotify.....	54
Growth Marketing.....	54
3 Engines of Growth.....	54
Scaling network effects.....	54
Types of network effects.....	54
Example: Instagram grew on external networks.....	54
Example: Paypal – using an existing community to grow.....	54
Negative network effects.....	54
Example: OKCupid – using matching algorithms.....	54
Customer Education Marketing.....	54
Data usage in Marketing.....	54
Information consumption preferences.....	54
Content creation and content distribution using 3rd party content platforms.....	54
User journeys are the new sales funnels.....	54
Website examples of businesses with a pipeline/platform overlap.....	54
Apple.....	54
SAP.....	54
Schneider Electric EcoStruxure.....	54
VUCA challenge.....	54
VUCA solution.....	54

Process management.....	54
Integrating Design Thinking, Lean Six Sigma, PDCA and Agile.....	54
Innovation and process management at Google.....	54

Definitions

There seems to be confusion about the usage/meaning of the word “platform”. People understand “platform” differently based on their professional background. Therefore, starting with definitions to distinguish 3 types of platforms in the platform economy context.

Marketplace platform

A marketplace in the context of a platform economy is a digital platform that **connects buyers and sellers, facilitating transactions between them**. These platforms provide a space where various vendors can offer their products or services to a wide audience, and consumers can browse, compare, and purchase from multiple sources in one place. Marketplaces can operate in various sectors, including retail, services, and even business-to-business (B2B) transactions.

The value of a marketplace in the platform economy lies in its ability to create a **network effect**, where the value of the platform increases as more users (both buyers and sellers) join.

Key benefits:

- **Efficiency:** Marketplaces streamline the buying and selling process by providing a centralized platform for transactions, reducing the time and effort required to find and connect with potential buyers or sellers.
- **Variety and Choice:** Consumers benefit from a wide range of options, as multiple vendors compete on the same platform, often leading to better prices and more diverse offerings.

- **Trust and Security:** Established marketplaces often implement measures to ensure secure transactions and build trust among users through reviews, ratings, and buyer protection policies.
- **Scalability:** For sellers, marketplaces offer an opportunity to reach a larger audience without the need for significant investment in marketing or infrastructure.
- **Data Insights:** Marketplaces can collect and analyze data on user behavior, preferences, and trends, providing valuable insights for both the platform operators and the sellers.

B2B examples: [Alibaba](#) (founded 1999), [ThomasNet](#) (transitioned to online in the late 1990's), [ABB Ability Marketplace™](#) (launched 2017), [SAP Store](#) (launched 2010)

Source: MS Copilot

Technology platform

A technology platform in the context of a platform economy is a **foundational digital infrastructure that enables the development, deployment, and management of applications and services**. These platforms provide the necessary tools, frameworks, and environments for developers to create software solutions. They often include APIs (Application Programming Interfaces), SDKs (Software Development Kits), and other resources that facilitate integration and interoperability with other systems.

The value of a technology platform in the platform economy lies in its ability to foster innovation, collaboration, and scalability.

Key benefits:

- **Innovation:** Technology platforms provide developers with the tools and resources needed to [create new and innovative applications](#), driving technological advancement and offering new solutions to market needs.
- **Collaboration:** By offering a common framework and set of standards, technology platforms enable developers from different organizations to collaborate and build complementary solutions, enhancing the overall ecosystem.
- **Scalability:** These platforms allow businesses to scale their operations efficiently by leveraging cloud infrastructure, automated deployment, and other scalable technologies.
- **Interoperability:** Technology platforms often support integration with various other systems and services, ensuring that applications can work seamlessly within a broader technological landscape.

- **Cost Efficiency:** By providing ready-made tools and infrastructure, technology platforms reduce the time and cost associated with developing and maintaining software solutions.

B2B examples: [SAP Business Technology Platform \(BTP\)](#) (launched 2012, initially known as SAP Cloud Platform), [Bosch IoT Suite](#) (first made available for third-party development in 2015), [Schneider Electric EcoStruxure](#) (launched 2016), [Amazon Web Services \(AWS\)](#) (launched 2006, 3rd part development support followed shortly after)

Source: MS Copilot

Development platform

A development platform is a **digital environment that provides the necessary tools, frameworks, and resources for developers to create, test, and deploy software applications**. In contrast to a technology platform, a development platform is a more specialized environment that includes integrated development environments (IDEs), version control systems, build automation tools, and testing frameworks.

A company with a technology platform does not necessarily have to provide a development platform for third-party development. Third-party developers can use their preferred development tools and environments to build applications on top of a technology platform.

For example, a developer can use Visual Studio or any other integrated development environment (IDE) to develop software that runs on the SAP Business Technology Platform (BTP). The technology platform provides the necessary infrastructure, services, and APIs, while the development platform (such as Visual Studio) provides the tools for coding, debugging, and testing the application. This flexibility allows developers to choose the development tools that best suit their needs while leveraging the capabilities of the underlying technology platform.

B2B examples: Microsoft Azure DevOps, GitHub, Mendix

Source: MS Copilot

What is a platform economy?

The platform economy is defined by Acquier et al. (2017) as “a set of initiatives that intermediate decentralized exchanges among peers through digital platforms.” Platforms have penetrated almost all sectors of modern economies and have adopted

very different forms and guiding motivations, but they usually share the following two characteristics: (1) they create value by facilitating transactions rather than through their own production and (2) they create strong network effects (i.e., their value rises with the number of users and suppliers joining) (Acquier et al., 2017). If the community-based economy is where the sharing economy has its deepest roots and the access-based economy is a necessary condition for the sharing economy to grow, the platform economy might be the boost that the sharing economy needed to scale up.

[Virginie Boutueil, in Urban Mobility and the Smartphone, 2019](#)

How economic value creation & processes changed with each industrial revolution

- **1. Industrial Revolution (17th/18th century)**

The primary force driving the first industrial revolution was the discovery of the steam engine. For the first time, labor could be replaced by mechanization, allowing machines to take over human labor. The key aspect of value creation during this period was **mechanization**. For example, before the 1st industrial revolution, highly skilled and paid crafts people finished woolen cloth at their homes. The so-called “cottage industry” was replaced by new mechanical frames in textiles mills (central location), producing cloth much faster than workers could by hand. Worst of all, no skill was required to use the frames and so textile workers in factories were paid low wages.

- **2. Industrial Revolution (19th century)**

The primary force was the discovery of **electricity**. This enabled the creation of supply chains that ensured complex tasks could be performed with high efficiency due to the division of labor. Over time, these supply chains were optimized and made more efficient. Which led to the famous example of Henry Ford whose Ford Motor Company introduced the Model T in 1908, of which 15.5 million were eventually sold in the United States. Producing so many Model Ts was possible because of the assembly-line process that Ford developed for his factories. Instead of hiring one or two specialists, Ford paid high wages to dozens of less-skilled workers to complete just one part of the process on multiple cars. Previously, a Model T took over 12 hours to build. The assembly line spit out the same automobile in just an hour and a half.

- **3. Industrial Revolution (20th century)**

The key shift was the invention of modern computers, which allowed programmable logic to be used for the first time to create value by **automating tasks**. Walmart succeeds and grows because of a relentless focus on efficiency

and transparency in its supply chain, even in packing pallets and stocking shelves. Demand forecasting and inventory-level prediction are obsessions. Walmart was the first company in the world to use barcodes on 100% of its products, way back in 1983. What's more, Walmart spent \$14 billion in 2021 on supply chain automation. This begins with automated warehouse management and control systems, which are software, and automated picking, sorting and putting systems, which are hardware. The software handles inventory control of the bewilderingly large assortment of Walmart's various online and in-store channels. It even forecasts labor needs to ensure that staffing levels exactly match the need for each shift, bringing in more hands when large deliveries and shipments are expected and keeping a smaller crew for quieter hours. Because all the relevant data in Walmart warehouses is transparent and interoperable, that labor can be provided by machines as well as people. High-speed robots are already loading and moving pallets, and automated sortation equipment is sorting products along conveyers and chutes. These technologies, together with data sharing, underpin automated consolidation centers, the first of which opened in 2019 in Colton, California, with more to come.

- **4. Industrial Revolution (21st century)**

The way value is created has changed significantly between the third and fourth industrial revolutions. In the fourth industrial revolution, we are not just using programmable logic; we are **digitizing every possible object, interaction, and workflow**. This data is used to create intelligence that not only improves processes but fundamentally reorganizes them. The most important aspect is the orchestration of networks. Companies that can connect and reorganize these networks to create value will be the most successful and powerful in this industrial revolution.

[The Fourth Industrial Revolution | Sangeet Paul Choudary, Seoul Summit 2016 \(21 min\)](#)

[Protesters: the Luddite Movement](#)

[How Henry Ford's assembly line revolutionized factory production](#)

[Wal-Mart's Successfully Integrated Supply Chain and the Necessity of Establishing the Triple-A supply Chain in the 21st century](#)

[Walmart's Supply Chain: A Detailed Look at How They Manage It](#)

[Platforms, the Sharing Economy, and Business Model Innovation \(1h17min\)](#)

The 3rd industrial revolution: Efficient pipelines eat inefficient pipelines

“Software is eating the world.” The slogan was originally used by Netscape founder Marc Andreessen in the title of a 2011 op-ed article in the Wall Street Journal to encapsulate how technology— particularly the Internet—has transformed the world of business. The story of Internet-enabled disruption as we’ve witnessed it so far has occurred in two main stages. In stage one, efficient pipelines ate inefficient pipelines. Most Internet applications during the 1990s involved the creation of highly efficient pipelines—online systems for distributing goods and services that outcompeted incumbent industries. Unlike traditional offline pipelines, online pipelines benefited from low marginal costs of distribution—sometimes as low as zero. This allowed them to target and serve large markets with much smaller investment. Traditional media companies were the first to feel the pinch. Newspapers were upended by the Internet’s ability to distribute news to a global audience without the traditional distribution costs (printing, shipping, retailing, delivery). An efficient pipeline had eaten an inefficient one. The unbundling of classifieds and other forms of advertising from editorial content then stripped the newspaper model of a crucial monetization mechanism, as the more efficient online method for delivering targeted advertising outcompeted the traditional ink-on-paper method.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

The 4th industrial revolution: Platforms eat efficient pipelines

Over time, many of the Internet distributors found ways to leverage granular data about consumer choice so as to serve their customers better than the traditional pipelines ever did. Business revolutions such as these embodied Andreessen’s vision of “software eating the world.” Today, having attained the status of a cliché, his vision needs an update: “Platforms are eating the world.” We’ve entered stage two of the disruption saga, in which platforms eat pipelines. [...] In the world of platforms, the Internet no longer acts merely as a distribution channel (a pipeline). It also acts as a creation infrastructure and a coordination mechanism. Platforms are leveraging this new capability to create entirely new business models. [...] Platforms, then, have economic advantages that enable them to grow faster than similar pipeline businesses. This phenomenon alone would lead to significant disruption of traditional industries, as platform businesses displace pipeline businesses at the top of the Fortune 500 rankings. But the era of platforms-eat-pipelines is disrupting businesses in many other ways as well. In particular, the rise of the world of platforms is reconfiguring the familiar business processes of value creation, value consumption, and quality control. [...] To

fuel the continued explosion of new sources of supply, platform businesses are steadily lowering barriers that might discourage producers.

Pipeline vs platform business model



In the past, firms owned the linear supply chain (pipeline) and tried to make it as efficient as possible. *“This is how we used to think about business. [...] We optimized these systems. And that is a lot of what our management, measurement accounting ... systems are all designed to make this flow very efficient. Now we are doing things that are very different. [...] Now we are looking at much more shared infrastructure and a much looser connection between the producer side and the core firm. Instead of controlling all of the resources in this value chain going from left to right, we’re allowing you to affiliate and we will provide you tools and infrastructure that will then make you efficient in your production and we’ll set up systems that do the matching and do the search. So that you can find valuable business partners, but we’re not going to own all of that. We’re moving into that facilitation role”.*

The Platform Shift

- Shift 1: From resource control to resource orchestration
- Shift 2: From internal optimization to external interaction
- Shift 3: From focus on customer value to focus on ecosystem value

The goal for a platform firm changes from control, entry barriers and differentiation to creating and facilitating valuable market exchanges.

The transition to Internet era firms (platform) resembles the transition to industrial era firms (pipeline) ... but for the opposite reason (see [network effects](#)). While giant industrial firms owned the supply side, giant Internet firms own the demand side ([from Porter's 5 forces to dynamic capabilities](#)). Microsoft has 85% market share in desktop office applications. Amazon has two thirds of all online book sales (print & digital). Any time platforms go up against products, the platform business models always win (see [economic impact](#)). Everything we teach in a business school takes on a new form (see [platform manifesto](#)): IT, Strategy, Operations ... all of them.

[Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry \(2018\)](#)

[The Rise of Digital Platforms - Prof Geoffrey Parker, MIT Initiative on the Digital Economy \(2016\)](#)

How value is created in pipeline vs. platform businesses

To understand the powerful forces that are being unleashed by the explosion of platform businesses, it helps to think about how value has long been created and transferred in most markets. The traditional system employed by most businesses is one we describe as a pipeline. By contrast with a platform, a pipeline is a business that employs a step-by-step arrangement for creating and transferring value, with producers at one end and consumers at the other. A firm first designs a product or service. Then the product is manufactured and offered for sale, or a system is put in place to deliver the service. Finally, a customer shows up and purchases the product or service. Because of its simple, single-track shape, we may also describe a pipeline business as a linear value chain.

In recent years, more and more businesses are shifting from the pipeline structure to the platform structure. In this shift, the simple pipeline arrangement is transformed into a

complex relationship in which producers, consumers, and the platform itself enter into a variable set of relationships. In the world of platforms, different types of users—some of them producers, some of them consumers, and some of them people who may play both roles at various times (see [Sony example at CES](#))—connect and conduct interactions with one another using the resources provided by the platform. In the process, they exchange, consume, and sometimes cocreate something of value. Rather than flowing in a straight line from producers to consumers, value may be created, changed, exchanged, and consumed in a variety of ways and places, all made possible by the connections that the platform facilitates.

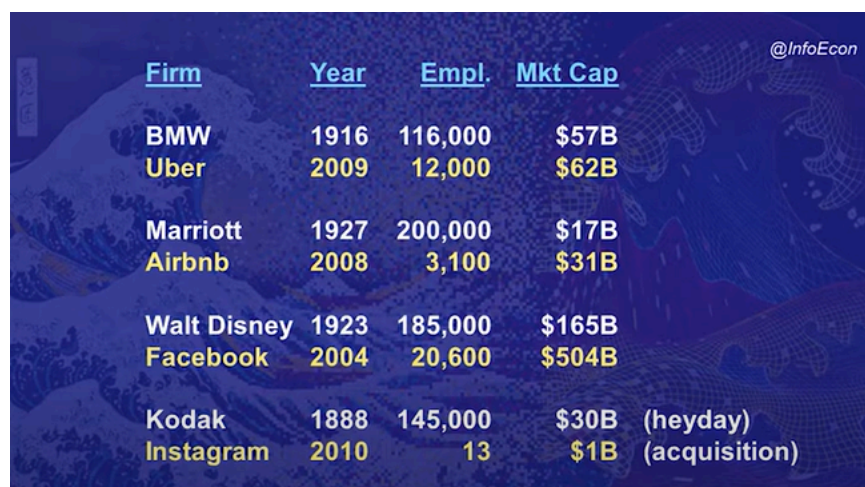
Every platform operates differently, attracts different kinds of users, and creates different forms of value, but these same basic elements can be recognized in every platform business. [...] The result is an exchange of value that is made possible by the platform itself. In itself, the shift from the traditional linear value chain to the complex value matrix of a platform may sound reasonably straightforward. But its implications are staggering. The spread of the platform model into one industry after another is causing a series of revolutionary changes in almost every aspect of business.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

[Platform Strategy and the Internet of Things](#) (2016)

Economic impact

The real shift happening is not moving from physical to digital. If we think that we come up with erroneous implications. Today, what we're seeing is not physical vs. digital but physical/digital hybrid models. The fundamental design of business itself is changing.



<u>Firm</u>	<u>Year</u>	<u>Empl.</u>	<u>Mkt Cap</u>
BMW	1916	116,000	\$57B
Uber	2009	12,000	\$62B
Marriott	1927	200,000	\$17B
Airbnb	2008	3,100	\$31B
Walt Disney	1923	185,000	\$165B
Facebook	2004	20,600	\$504B
Kodak	1888	145,000	\$30B (heyday)
Instagram	2010	13	\$1B (acquisition)

@InfoEcon

In this new stage of disruption, platforms enjoy two significant economic advantages over pipelines. One of these advantages is superior marginal economics of production and distribution. As we've noted, when hotel chains like Hilton and Sheraton want to expand, they build new rooms and employ thousands of staff. By contrast, Airbnb expands with near-zero marginal costs, since its cost for adding an additional room to its network listings is minimal. A platform's ability to scale rapidly is further enhanced by network effects. When positive network effects kick in, higher production leads to higher consumption, and vice versa. More freelancers participating on Upwork make it more attractive to companies looking to hire, which in turn attracts more freelancers; a wider array of merchants on Etsy attracts more customers, who in turn attract more merchants. A virtuous feedback loop is set in motion, fueling the growth of the platform at minimal cost. Leveraging network effects, platforms are able to build open electronic ecosystems embracing hundreds, thousands, or millions of remote participants. Such ecosystems can be larger than most pipeline-based organizations and can have access to more resources than a traditional pipeline company can command. As a result, [the value created in such an ecosystem can be much larger than the value created in a comparable traditional organization](#). Therefore, firms that continue to compete on the basis of resources that are owned internally are increasingly finding it difficult to compete with platforms.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

[Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry \(2018\)](#)

[The Fourth Industrial Revolution | Sangeet Paul Choudary. Seoul Summit 2016 \(21 min\)](#)

Network effects

Network effects refer to the impact that the number of users of a platform has on the value created for each user. Positive network effects refer to the ability of a large, well-managed platform community to produce significant value for each user of the platform. Negative network effects refer to the possibility that the growth in numbers of a poorly managed platform community can reduce the value produced for each user.

The network effect represents a new economic phenomenon, driven by technological innovation. In the twentieth-century industrial era, giant monopolies were created based on supply economies of scale. These are driven by production efficiencies, which reduce the unit cost of creating a product or service as the quantities produced increase. These supply economies of scale can give the largest company in an industrial economy a cost advantage that is extremely difficult for competitors to overcome.

In the twenty-first-century Internet era, comparable monopolies are being created by [demand economies of scale](#) (a term used by the two experts largely responsible for popularizing the concept of network effects, Hal Varian, the chief economist at Google, and business professor Carl Shapiro). By contrast with supply economies of scale, demand economies of scale take advantage of technological improvements on the demand side—the other half of the profit equation from the production side. Demand economies of scale are driven by efficiencies in social networks, demand aggregation, app development, and other phenomena that make bigger networks more valuable to their users. They can give the largest company in a platform market a network effect advantage that is extremely difficult for competitors to overcome.

Demand economies of scale are the fundamental source of positive network effects, and thus the chief drivers of economic value in today's world. This is not to say that supply economies of scale no longer matter; of course they do. But demand economies of scale, in the form of network effects, have become the most important differentiating factor.

[Metcalfe's law](#) is a useful way of encapsulating how network effects create value for those who participate in a network as well as for those who own or manage the network. Robert Metcalfe, coinventor of Ethernet and founder of 3Com, pointed out that the value of a telephone network grows nonlinearly as the number of subscribers to the network increases, making more connections among subscribers possible.

When there's only one node in a network, no connections are possible. An MIT professor we know likes to joke that the prize for "greatest salesperson in history" should go to whoever sold the first telephone. Arguably, it had zero value, because when there's only one telephone in the world, you can't call anyone. But as more people buy telephones, the value grows. With two telephones, one connection is possible. With four telephones, six. With twelve, sixty-six. And with 100 telephones, there are 4,950 connections. This is known as nonlinear or convex growth, and it is precisely the characteristic growth pattern seen in companies like Microsoft in the 1990s, Apple and Facebook today, and Uber tomorrow. (Working in reverse, it explains the convex collapse of Blackberry in the 2000s: as users began to flee the Blackberry platform, the loss of network nodes caused the value of the network itself to plummet, encouraging still more people to abandon Blackberry for other devices.)

Major economic consequences follow from this pattern. Growth via network effects leads to market expansion. New buyers enter the market, attracted by the growing number of friends who are part of the network. If prices also fall—as they often do when

technology matures, and production quantities increase—then network effects work together with more attractive pricing to drive massive market adoption.

Network effects are creating the giants of the twenty-first century. Google and Facebook each touch more than one-seventh of the world's population. In the world of network effects, ecosystems of users are the new source of competitive advantage and market dominance.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Monetization of network effects

Meeting the monetization challenge must begin with an analysis of the value created on the platform. Traditional non-platform businesses—pipelines—deliver value to their customers in the form of a product or service. They may charge for ownership of the product, as Whirlpool does when it sells a dishwasher, or for utilization of the product, as [GE Aviation](#) charges for installation and regular servicing of its aircraft engines.

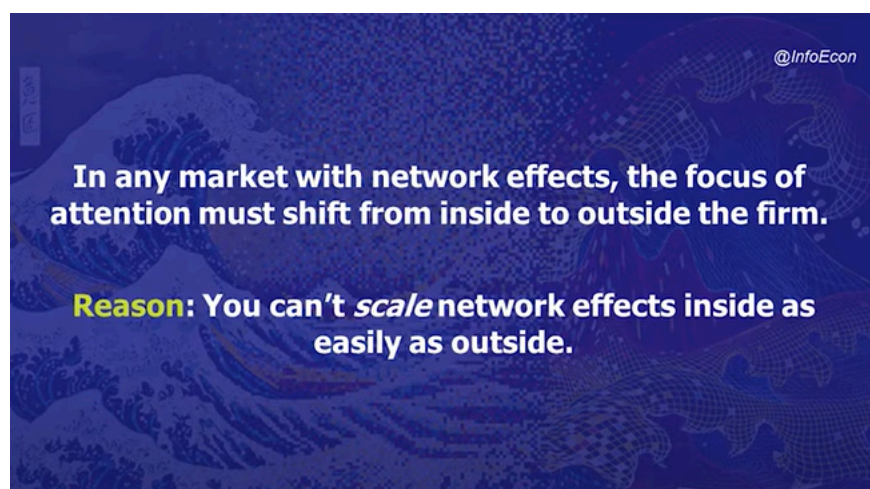
“Here’s the idea,” one of the founders explained to Marshall. “Our new platform Ad World will provide a listing service for firms to find ad agencies. We’ll make it easy for companies that are ready to begin new advertising campaigns to post requests for bids, and for ad agencies to post proposals and offers that companies can look at and respond to. It’s just like 99designs, which lets graphic artists link up with consumers who want help with artistic projects, only in the B2B space instead of the B2C space. [...] Here’s what we want to know,” the founder replied. “We’re pretty sure Ad World is going to provide value to our users and attract a fair amount of interest. But we’re wondering how to generate revenues from it. Should we charge the ad agencies to join the platform and put up their profiles? Should we charge the firms that are seeking services? Or should we charge for individual project listings? Or maybe for all three?”

These founders shouldn’t charge either side to be listed on their platform. Doing so would put terrible friction on entry into the ecosystem, discouraging many potential participants from becoming users. To charge for posting a deal simply means people post fewer deals. That’s bad. It reduces potential interaction volume, not to mention realized interactions. As a result, it also reduces the volume of data available to the platform—data desperately needed to enable the platform to forge powerful matches between consumers and producers. [...]

In fact, rather than charging users to join the platform, the founders should be subsidizing their participation—perhaps by providing tools and services to make it easy, fast, and effective for them to complete their profiles. [...]

How, then, could the founders monetize their platform model? The answer: They can charge users for the value they accrue from the ecosystem, but the charge should be levied on deal completion, not at the time of listing. They would make it possible for firms to post a deal risk-free by charging a fee only once the firms get what they need.

The fee becomes performance-based, and it feels negligible because it simply skims off a small fraction of a transaction that's occurring anyway. What's more, the best strategy may be one the founders didn't even consider. **Why not charge ad agencies for a service that helps them do a postmortem to discover why they lost a deal?** Not only would this fee create no friction on a deal, but it would reflect the value of the feedback provided; it would be recurring revenue rather than one-time revenue; and it could help ad agencies improve the quality of their offerings, thereby encouraging a rise in the value of interactions over time.



[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

[Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry \(2018\)](#)

Organizational impact

Platform businesses, then, are disrupting the traditional business landscape in a number of ways—not only by displacing some of the world's biggest incumbent firms, but also by transforming familiar business processes like value creation and consumer behavior as well as altering the structure of major industries.

What can incumbents do to respond? Are entrenched companies that operate familiar pipeline businesses doomed to capitulate as platforms reshape and ultimately take over their industries? Not necessarily. But if incumbents hope to fight the forces of platform disruption, they'll need to reevaluate their existing business models. For example, they'll

need to scrutinize all their transaction costs—that is, the money they spend on processes such as [marketing](#), sales, product delivery, and customer service—and imagine how those [costs might be reduced](#) or eliminated in a more seamlessly connected world. They'll also need to examine the entire universe of individuals and organizations they currently interact with and envision new ways of networking them so as to create new forms of value.

They'll need to ask questions such as:

- Which processes that we currently manage in-house can be delegated to outside partners, whether suppliers or customers?
- How can we [empower outside partners](#) to create products and services that will generate new forms of value for our existing customers?
- Are there ways we can network with current competitors to produce valuable new services for customers?
- How can the value of the goods and services we currently provide be enhanced through new data streams, interpersonal connections, and curation tools?

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Dynamic capabilities

Definition of Dynamic Capabilities:

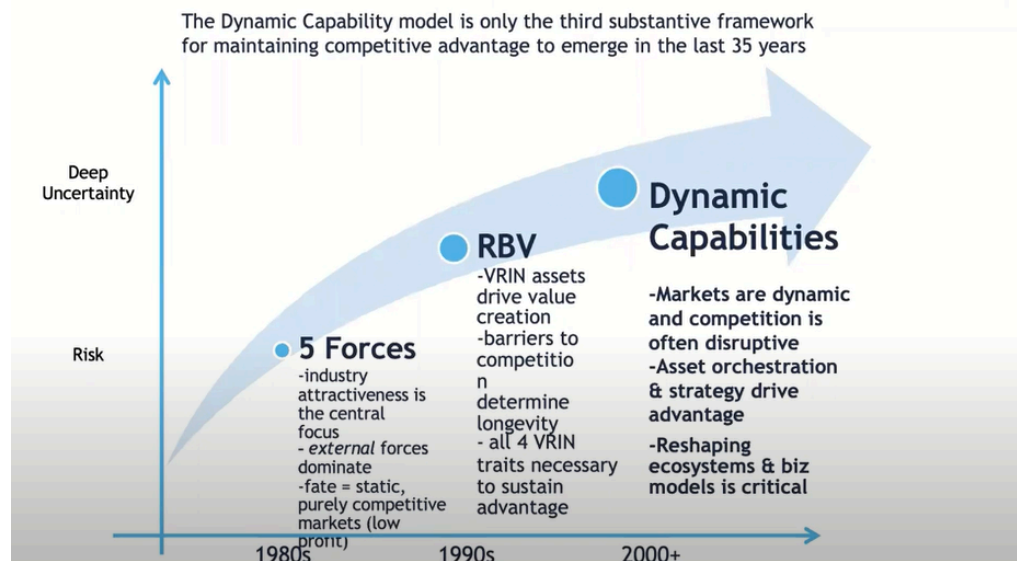
"The ability of an organization and its management to integrate, build, and reconfigure internal and external competences to address rapidly changing environments"

(Teece, Pisano, and Shuen, 1997: 516)

The platform economy (and other factors) requires us to look at what competitive advantage means differently. Within a platform ecosystem, [Porter's 5 forces](#) (e.g.

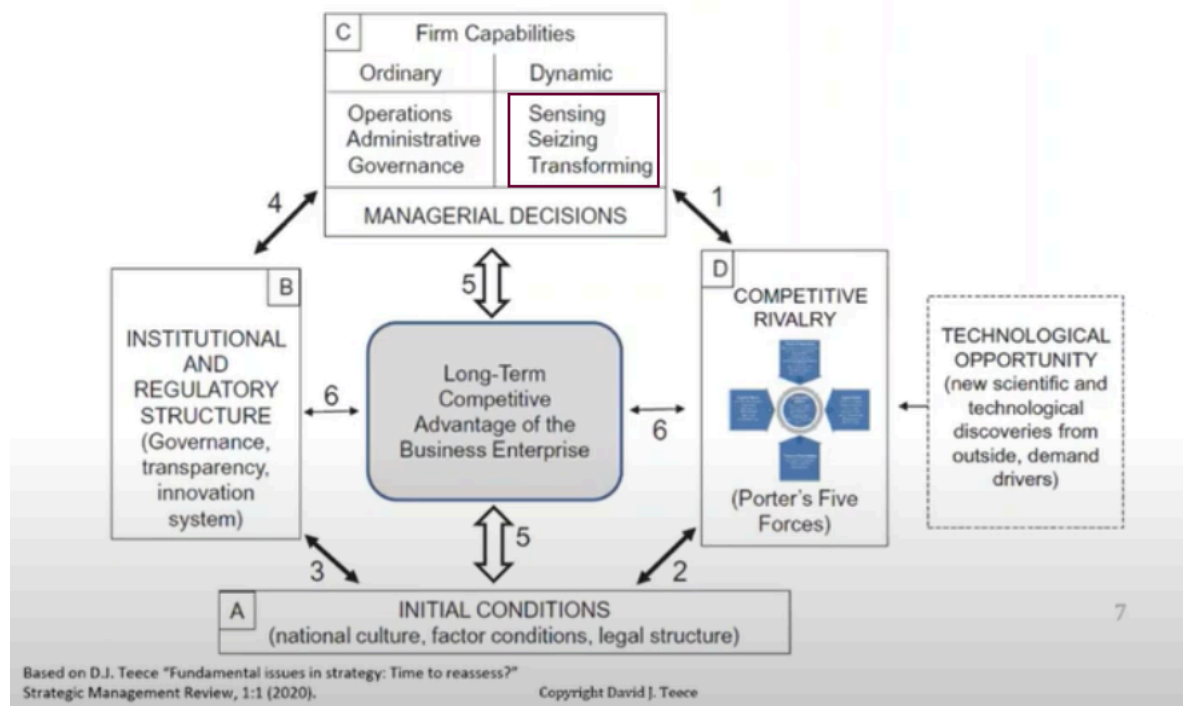
barriers to entering a market = friction) are no longer a competitive advantage but prevent innovation. However, the 5 forces are still relevant when it comes to competition between platform ecosystems. There is a place for understanding basic rivalrous nature, but we have to reframe it away from industry towards ecosystems.

The Evolution of Strategic Management and “Research Based” Thinking



"It is important to understand initial conditions, particularly with the rise of China and the decline of rule of law systems of governance. There's a different kind of competition [...] I think you're going to see in Europe and the United States greater coordination between firm strategies because that is going to be necessary to compete with China. The EU antitrust decision to strike down the merger between Siemens and Alstom was blind to the nature of competition from China."

THE (INTEGRATED) FOUNDATIONS OF FIRM-LEVEL COMPETITIVE ADVANTAGE



Dynamic capabilities can be thought of as falling in three categories:

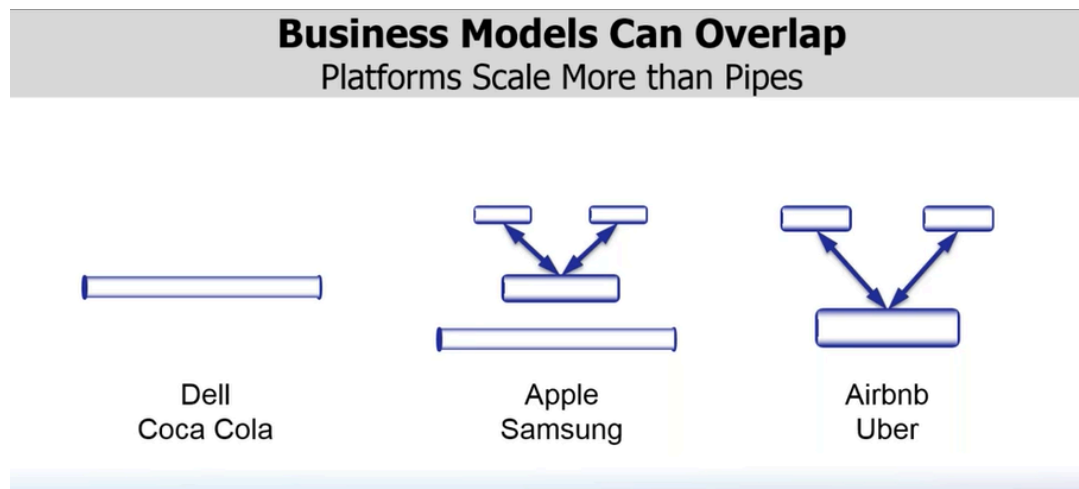


[Corporate Strategy Masterclass: The Evolution of Corporate Strategy \(2021\)](#)

[Dynamic Capabilities with Dr. David Teece \(2022\)](#)

Is it one (platform) or the other (pipeline)?

Short answer: No. There are asset-free platform companies like Uber or AirBnB. There are companies who will/might remain pipeline-only businesses who might join platform ecosystems as suppliers. And there are companies who are pipeline businesses and build a platform on top of their pipeline business.



Not all markets will be disrupted. And even if a market is disrupted, there might still be undisrupted niches (those actors might join a platform business as a supplier). Even today, some people prefer to design their kitchen table themselves, and have it manually produced by a carpenter.

Platform Disruption/Transformation Checklist	
Current Transformation	Eventual Transformation
High proportion of value from information (e.g. news, video) <i>Provides NFX spillover</i>	High proportion of value from physical assets (e.g. mining, construction) <i>Prevents NFX sharing</i>
Precise, fine-grained, modular output (e.g. tweet, ride, stay, retail) <i>Allows 3rd party supply / curation</i>	Highly complex products / services (e.g. Airbus and Boeing aircraft) <i>Prevents 3rd party supply</i>
Lightly regulated, fault tolerant (e.g. videos, apps, tweets) <i>Safe for experimentation</i>	Highly regulated, fault intolerant (e.g. medical, energy) <i>Dangerous experiments, malice</i>
Spare Capacity (e.g. Uber, Airbnb, loans, after markets) <i>Utilization market, Decouple asset from use</i>	High utilization (e.g. surgeons, phones, perishables) <i>Own not rent</i>

Source: Platform Revolution, 2016

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

[Platform Strategy and the Internet of Things](#) (2016)

[How Platforms Change Structure and Strategy](#) Marshall Van Alstyne (2016)

Geoff Parker, Platform Economy Summit, 2018 (24 minutes)

Beyond the marketplace: The competitive advantage of technology platforms (speed of innovation)

Does Openness Work? @InfoEcon

"We tried to create every feature in the world and said, 'O.K., we can do it, why should we let a third party do it?'" says (MySpace cofounder) DeWolfe.

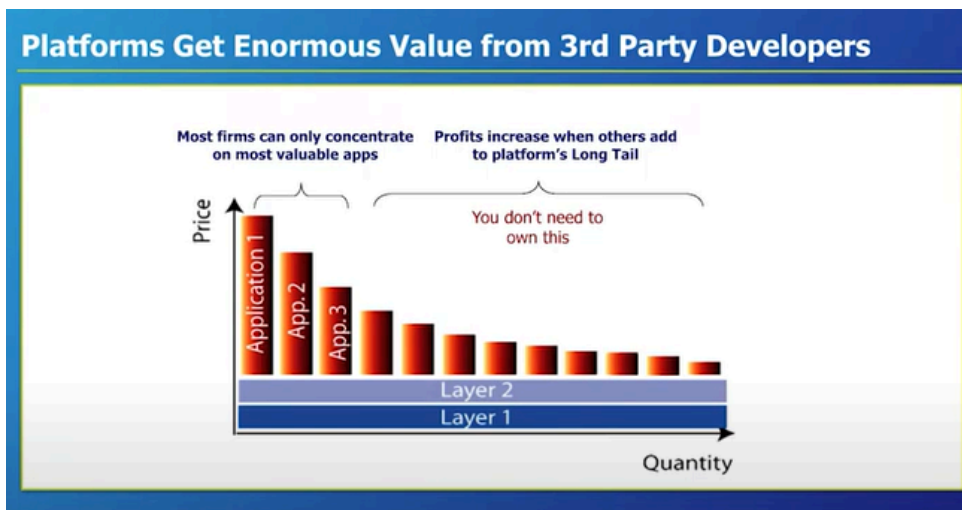
"We should have picked 5 to 10 key features that we totally focused on and let other people innovate on everything else."

The Rise & Ignominious Fall of MySpace – Business Week 2011

Twitter: @InfoEcon :: mva@bu.edu :: PlatformEconomics.com © 2018 Parker, Van Alstyne 49

"Same mistake Apple made in the 1980s and 90s. They kept too close. They tried to control it all to themselves. You need openness for people you don't know to bring you ideas you don't have. And you have to reward them for doing so. You have to enable it, you have to reward it. Platforms have these rules of governance. The mistake that technologists usually make is simply to throw it open and hope someone will come and build on it. You have to reward them. Thank them for giving you ideas that you didn't have. "

"If this is Microsoft. You own Word, Powerpoint and Excel and let the ecosystem build all this other stuff because it brings you all the additional value you didn't think about. The last app, trivial. You don't really care about that individual app, but you do really care about millions of trivial apps. [...] You want that long-tail innovation that 3rd parties are bringing that keeps you ahead of the curve."



A marketplace can be a technology platform. For example, [Hilton and Uber](#) have expanded their partnership by integrating their mobile apps to enhance the travel experience. Users can now order an Uber directly from the Honors app, while Uber riders can view hotel stay details, check in, choose their room, and request a Digital Key — all while en route to the hotel.

[Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry](#)

[Uber API Integration: A Complete Guide with 7 Benefits & Examples](#)

How does this work in B2B?

The most familiar platform examples—Airbnb, Uber, Amazon— come from the business-to-consumer (B2C) arena. How do you convert a product to a platform in the business-to-business (B2B) arena? Many corporations own massive fixed assets like power generation plants, magnetic resonance imaging (MRI) machines, or tracts of farmland. How do you build platforms around those?

1. **De-link ownership of the physical asset from the value it creates.** This allows the use of the asset to be independently traded and applied to its best use—that is, the use that creates the greatest economic value—rather than being restricted to uses specific to the owner. As a result, efficiency and value rise dramatically.

Example: Health care equipment

De-linking assets from value also allows expensive health care equipment such as MRI machines (each costing \$3–5 million) to be used more efficiently. A single hospital may use just 40 to 50 percent of its own MRI capacity. Solution: time-slice the usage and create a market for slices among other hospitals and

small clinics that cannot afford their own machines. Separating the asset from the value it creates can drive the utilization rate to 70 or even 90 percent, producing incremental revenue for the machine's owner.

--> Sales to Honeywell and Siemens go down, but the marketplace goes up in that context ([Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry](#), min 33:50)

This might be [Rayus Radiology](#)

2. Across numerous industries, platforms have repeatedly **re-intermediated markets**, introducing new kinds of middlemen rather than simply eliminating layers of market participants. [...] Networked platforms serve as more efficient intermediaries owing to their ability to use market mediating mechanisms that scale. While traditional intermediaries relied on manual efforts, platform intermediaries rely on algorithms and social feedback, both of which scale quickly and efficiently. Moreover, their ability to gather data over time and use it to make the system more intelligent allows platforms to scale their intermediation in the market in a manner that was impossible for traditional middlemen.

Example: Google Adwords

Small businesses run advertising campaigns without using traditional ad agencies or media channels by relying on Google's AdWords platform. This, in turn, has led to the rise of an entire new range of intermediary agencies in Asia that manage AdWords campaigns for a fraction of the traditional price. Thus, while platforms displace large and inefficient intermediaries, they empower small and nimble service providers who leverage the platform to provide services to end users.

3. Platforms are also creating new efficiencies by **aggregating unorganized markets**. Market aggregation is the process whereby platforms provide centralized markets to serve widely dispersed individuals and organizations. Market aggregation provides information and power to platform users who formerly engaged in interactions in a haphazard fashion, often without access to reliable or up-to-date market data.

Examples: Bus transportation in India & Upwork

Different bus fleets operate on interstate, intrastate, and other routes. Many different types of bus exist, and pricing is extremely variable. Because the industry is so fragmented and unorganized, consumer search costs and decision overhead are high. Now a platform business called redBus is aggregating information from all Indian bus operators in a central plug-and-play infrastructure.

The result is quicker, easier, and cheaper decision-making for consumers, and, in the long run, a healthier transportation market for India. Service platforms like Upwork bring thousands of skilled professionals under a single roof, making it easy for potential employers to evaluate, compare, and hire them.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Four types of B2B platforms

Businesses cannot fit their B2B efforts into a standardized model because there are four different B2B platform types:

5. **Technology enablement platforms**, which harness technology building blocks like the Internet of Things or process automation and provide an opportunity to innovate on top of them.

One example is Dassault Systèmes' 3DEXperience platform, which is aimed at helping companies design, engineer, and manufacture new products.

6. **Intelligent products and services**, which enrich existing products and assets with new services and digital functions.

For example, the Danfoss Alsense platform lets customers monitor their refrigeration assets anywhere in the field using cellular networks.

7. **Data aggregation and collaboration platforms**, which collect and aggregate data from multiple sources and provide tools to enable new analytical services and collaboration opportunities.

The Siemens SiGreen emissions management tool measures and reduces product-related CO2 emissions across the supply chain.

8. **Marketplaces** that facilitate supply and demand transactions for goods and services.

For example, the European marketplace CheMondis is a user-friendly option for connecting buyers and suppliers of chemicals, with the goal of expanding sourcing networks.

[Why B2B platforms struggle to succeed \(2023\)](#)

Example: Energy market aggregation

Sidenote: It sounds as what Rifkin refers to as the 3rd industrial revolution is what Geoffrey G. Parker, Marshall W. Van Alstyne & Sangeet Paul Choudary (Platform Revolution) refer to as the 4th industrial revolution.

Today, the Internet is converging with renewable energies and becoming the communication media to manage a distributed and collaborative green economy. In the future, hundreds of millions of people will produce their own green energy in their homes, offices, and factories and share it with each other on an "energy internet," just like we now create and share information online. The creation of a renewable energy regime, loaded by buildings, partially stored in the form of hydrogen, distributed via an energy internet, and connected to plug-in zero emission transport, establishes the essential five pillars of a Third Industrial Revolution infrastructure.

When we look back in history the great economic revolutions occur when new energy regimes emerge. New energy regimes make possible more complex civilizations, but the complexity then requires new communication revolutions that are agile enough to manage the new energy regimes. [...]

What's happening now in Europe is that this distributed collaborative laterally scaled communication revolution (the internet) is just now beginning to merge with a new energy regime: distributed energies which have to be organized collaboratively and scaled at lateral peer-to-peer power. It's a perfect match of communication and energy. [...]

How do we collect energies that are distributed? Sun, wind, geothermal heat, waves and tides. If distributed energies are everywhere, why don't we collect them everywhere? So, we collect them where there's buildings now. In the EU, we have a hundred and ninety-one million buildings. Homes, offices, factories, and the goal is to convert every single existing building in the EU to your own personal green micro power plant. This is where the internet communication revolution converges with the distributed renewable energies. It's the nervous system for this new economic paradigm.

[Saying goodbye to Adam Smith at the Dawn of the Third Industrial Revolution](#) (Guardian, 2011)

[Jeremy Rifkin speaks to "Living Energy" about the Third Industrial Revolution](#) (Siemens Knowledge Hub, 2013, 12 min)

Why is the industrial space lagging?

This happened before. During the [2nd Industrial Revolution](#) (electrification) the whole way of doing manufacturing had to be reconfigured, because it was not enough just to install electric motors and scrap the old technology – steam engines ([digital thread story](#)). In addition to the reorganization of the pipeline business, the industrial space is not as easily disrupted because of the high value from physical assets, high regulation,

utilization and complexity. Another explanation might be the [innovator's dilemma](#) which is often misinterpreted as incumbents failing to develop disruptive technologies or embrace them due to the inability of the organization to adapt operationally or technologically. The problem is they fail to value new innovations properly because incumbents attempt to apply them to their existing customers and product architectures — or value networks. Often new technologies are too new and weak for the more advanced and mature value networks that incumbents operate.

And what about hardware?

“To me, at the end of the day, machine-to-machine (M2M) is really about data. You’re getting data off devices or sensors that have been hard to get, but once it’s in the cloud, it’s data. So, all the data tools, if it is big data or any kind of business intelligence software, all that stuff is applicable to it. So, I think just the same tools and technologies, like Google or Facebook, really do a lot of analysis on their users, and the amount of data that they have, they’re very applicable into the M2M space. [...]

When looking at key players, their positioning, and potential profit pools in the near future, it is important to understand each player's unique position in the market, potential external effects on the supply and demand side, new players that are entering the market, and potential disruption that may change the overall industry dynamic or allow for alternative approaches and solutions that have not been feasible before. Each of the players in the value chain has a chance to improve their competitive positioning by building a unique offering. Commoditization will happen where there is lots of competition and no differentiation. Applying these principles to the M2M market, one can assume the largest profit pools are expected in the hardware solution design, and data analysis spaces. Why should that be? Because although there’s a common perception that it’s hard to make money with hardware, we believe that in M2M there are still a lot of opportunities to reap benefits from hardware. First of all, complexity is one of the key reasons the market has not taken off as expected. Most complexity in M2M today is in devices, but if one could remove this complexity, significant value could be unleashed for everyone in the ecosystem. [...]

On the other side of the supply chain are software and application platforms. This market is still highly fragmented and allows for high margins for system integrators, making many business cases less attractive than they could be. This market always reminds us of the mobile phone industry in the early days and the fragmentation issues that there were for a number of years. If a couple of players are able to design application platforms that allow for easy and cost-effective development and deployment of application, these platforms would be able to benefit from massive

network effects, like the ones we have seen around mobile platforms starting in 2006. [...]

Finally, there is another important factor, which is the [network effect](#). This tends to show exponential growth once you have a critical mass of intelligent items and connected items that are talking to one another ([Metcalfe's Law](#)). This law is applicable not only to telecommunications and social networks, but also to connected things, like M2M nodes. Assuming such a network effect will take place, one can assume it will create a huge value for the ecosystem.”

[The Silent Intelligence, Daniel Kellmereit, Daniel Obodovski, 2013](#)

Example: GE Aviation

General Electric (GE) has undergone this massive transformation. It's gone from being an industrial equipment company into being a software company. When GE sells aircraft engines, for example, they sell them bundled with a maintenance contract. If you are in charge of operations at GE what is your main problem? What's the number 1 cost that you have to bear? Your maintenance costs. Because the maintenance cost a customer pays doesn't vary based on the amount of maintenance needed. GE would send out these maintenance teams to visit the engines periodically. Half the time the maintenance people get there, they look at the engine and everything is fine. The other half of the time they get there, and the engine is falling apart, and they think, if we had gotten here a week earlier, we could have repaired this thing.

So, what would you need to know to arrive at exactly the right time? You'd like to be able to track the degradation of the engine in real time. You would put sensors all through the engine. General Electric now has dozens of sensors installed in every engine which pick up pressure, temperature, wind speed ... which gives them the ability to monitor the health of the engine. The amount of information generated by these engines is astronomical. About 2 terabytes an hour. What they can do with that information is reduce these maintenance costs to a very low level.

But what if the Marketing group gets hold of the data Operations collects? If I am marketing and I know how my customer uses my product I can start charging them differentially. I can say, if you fly low altitude vs. high altitude, desert vs. ice environment that has an impact on maintenance. I can vary the price based on how you use the product. Now, we're talking about transforming business. Because now we're not talking about selling engines, but engines as a service. But I don't just sell services like access to the engine, I can now bundle access to the engine with advice on how to use this

engine better, like I have noticed you're using this engine in a way that is not fuel efficient. Maybe if you do it this way, it'll be more fuel efficient.

[Platforms, the Sharing Economy, and Business Model Innovation](#)

What other business platforms exist in the industrial space?

1. GE's Predix

Description: Predix is an industrial IoT platform developed by General Electric (GE) to connect machines, data, and people for industrial applications.

Launch Year: 2015

Components: Technology Platform, Ecosystem, Marketplace

2. Bosch IoT Suite

Description: Bosch IoT Suite is a comprehensive platform for IoT solutions, offering services for device management, data management, and analytics.

Launch Year: 2016

Components: Technology Platform, Ecosystem

3. [Schneider Electric's EcoStruxure](#)

Description: EcoStruxure is an IoT-enabled, plug-and-play, open architecture platform developed by Schneider Electric for various industries, including energy, buildings, and data centers.

Launch Year: 2016

Components: Technology Platform, Ecosystem, Marketplace

4. ABB Ability™

Description: ABB Ability™ is a unified, cross-industry digital platform that provides a range of digital solutions and services for industrial automation and electrification. **Launch Year:** 2017

Components: Technology Platform, Ecosystem, Marketplace

5. Dassault's 3DEXperience

Description: 3DEXperience is a business experience platform developed by Dassault Systèmes, offering software solutions for design, engineering, and manufacturing in a collaborative environment.

Launch Year: 2012

Components: Technology Platform, Ecosystem, Marketplace

Source: MS Copilot

The Platform Manifesto

As with the [Agile Manifesto](#), prioritizing the left side doesn't mean ignoring the right side.

1. The ecosystem is the new warehouse
2. The ecosystem is also the new supply chain
3. The network effect is the new driver for scale
4. Data is the new dollar
5. Community management is the new human resources management
6. Liquidity management is the new inventory control
7. Curation and reputation are the new quality control
8. User journeys are the new sales funnels
9. Distribution is the new destination
10. Behavior design is the new loyalty program
11. Data science is the new business process optimization
12. Social feedback is the new sales commission
13. Algorithms are the new decision makers
14. Real-time customization is the new market research
15. Plug-and-play is the new business development
16. The invisible hand is the new iron fist

[Sangeet Paul Choudary. Platform Scale \(2015\)](#)

What is an ecosystem in the platform economy?

A Digital Ecosystem is a **socio-technical system in which companies and people cooperate that are independent but expect to gain mutual benefits from participating**. A Digital Ecosystem has a digital platform at its center that supports this cooperation particularly well via ecosystem services.

The overall benefit of a Digital Ecosystem thus results from the combination of the digital brokering platform and a large set of partners who participate in the Digital Ecosystem for mutual benefit and whose interactions via the platform lead to network effects.

[Fraunhofer Institute for Experimental Software Engineering IESE](#)

In Business research, propositions to rethink business environments using a social network or ecosystem approach have been fashioned. An ecosystem in this context is comprised of **diverse organizational and individual actors that contribute to the company's business**. These actors may be diverse in types and situated in different domains but are interdependent. Business ecosystem research usually reflects either of the two foci: **affiliation or structure**. The former places **emphasis on network diversification and the rise of interdependencies and highlights how the ecosystem dissolves traditional organizational boundaries**. The latter focuses on investigating **alternative value proposition mechanisms, which create (endogenous) boundary of the**

relevant ecosystem. Earlier research on business ecosystems emphasizes the key role of central actors in providing and coordinating a stable, accessible foundation for collaboration and ecosystem improvement.

[The platform economy and futures of market societies: Salient tensions in ecosystem evolution](#)

An ecosystem is a **system of interacting actors who interact and coordinate towards a value creating outcome**. A platform, on the other hand, is an organizing mechanism that organizes these interactions to achieve the above value creation outcome. Together, they constitute the platform ecosystem. **Ecosystems can exist without platforms**. For example, contracts and standards are alternate coordination mechanisms that may be used to manage ecosystems. Before the rise of platforms, most business ecosystems were coordinated through contracts.

[Platform ecosystems: Why they matter, the good and the bad – A deep-dive](#)

An ecosystem: Organizations of organizations

Platform-based ecosystems are a vital source of dynamism and innovation for many technologies, products, and services in the global economy. By leveraging modularity and specialization, platforms have spurred innovation and increased efficiency in diverse sectors such as computers, videogames, mobile phones, automobiles, payment systems, e-commerce, government, and more.

A defining feature of platform ecosystems is the interdependence between a stable core or “platform” that interfaces with a dynamic and heterogeneous set of complementary components to generate a stream of derivative products. Platform ecosystems (PEs) can be understood as economic communities of interconnected organizations and individuals built around a core setting or platform.

Drawing on an ecosystem perspective, PEs are complex networks formed by organizations that interact and share resources to provide customer-oriented solutions. Consequently, these systems create shared spaces for participants to collaborate and co-create products and services, in ways that are perceived as valuable by the economic communities. As such, the success of platforms hinges on cooperation, coordination, and integration across a diverse and often very large array of organizational units and agents, some of whom face conflicting incentives or are direct rivals.

In a traditional hierarchical organization, authority is based on ownership of the assets necessary for production, and it deploys that authority through employment contracts and the (delegated) power of managers to exclude employees who do not comply. In

platform ecosystems, in contrast, the source of authority and power of a platform “hub” largely arises from control over the technological architecture at the center of the ecosystem, and relational centrality among a number of players for whom coordination offers the promise of greater rents.

Another key distinction between platform ecosystems and traditional organizations is the nature of motivation and incentives. Whereas a traditional organization usually provides low powered incentives by directly paying employees recurring wages for their services, in platform ecosystems the incentives and compensation are indirect and high-powered, as they involve business transactions among the participants themselves. Platforms typically enable transactions or interactions among participants that allow them to create value for one another and also capture some of that value.

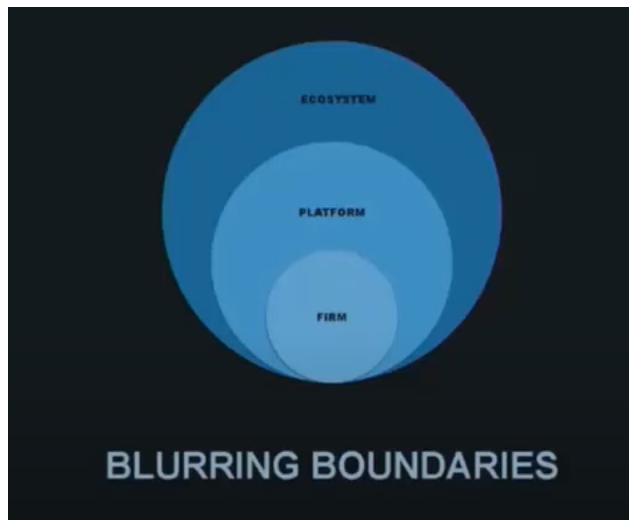
TABLE 1 Meta-organizational features and platform strategies

Meta-organizational features	Strategic dimensions		
	Platform entry	Between-platforms competition	Within-platform competition
Sources of power and authority	Technological and/or relational hub to facilitate coordination; enables modularity and loose coupling for participant autonomy and flexibility	Architectural control of the ecosystem to capture more value	Power to influence the distribution of profits across stakeholders
Sources of motivation for participation	Incentives to attract participants under uncertainty	Incentives for users, and participants to offer high quality complements that create value in the platform ecosystem; incentives to encourage loyalty	Incentives to balance cooperation and competition with participants
Modes of governance and coordination	Use technology to lower coordination costs and enable search and matching; create technology-based trust and security	Balance rules and decision rights with openness; balance coordination and participant contributions; balance cooperation and competition with other platforms	(De)centralize decision rights to manage cooperation and competition among complementors

[Platform ecosystems as meta-organizations: Implications for platform strategies \(2020\)](#)

[Unveiling the core elements of platform ecosystem development: a systemic lens for value co-creation in small and medium enterprises and orchestrators \(2024\)](#)

What does this look like?



What we are seeing is that the boundary of your firm is becoming more blurred. [...] We've seen this shift happening for some time. For the first time, we have a mechanism by which all of these ecosystem players can be orchestrated together off a central platform, and that is the view I have for all [incumbent organizations](#) when they move from a firm only model to a firm plus ecosystem model. When you move in this direction the first and most important thing you need to do is to integrate your enterprise, so you have one view of the ecosystem. If every business unit within your enterprise has a different data set about your suppliers, has a different identity of your customer you cannot work with the ecosystem in a way that is efficient, in a way that leverages the scale that a platform provides.

- The first step is to find the infrastructure that enables you to [end-to-end integration of your current enterprise](#), getting different pieces of infrastructure, different pieces of software from different vendors and digitizing different parts of your functions is not going to help run a platform model. You need an **end-to-end infrastructure** provider to integrate your enterprise.
- Then you need to start thinking about what should be **open**, [because opening everything is not good](#). You need to open in a way that innovation increases. More comes in from outside from your suppliers, but at the same time you close the critical points of control in your ecosystem.
- Once you start opening up, you start seeing **network effects**. The more people come onto your platform; the more other people benefit from all of these people in the ecosystem. The more producers come on board; the more suppliers come on board; the more consumers start benefiting. The more consumers come on

board; the more suppliers will come. [...] The moment you [digitize the source of value](#) or the consumer decision-making factor, every company, every industry is a tech industry. But it is not just about quantity of interactions, it is also about quality of interactions.

- So, the next thing you need to do is govern the ecosystem. Not everybody should come on board. Some people should not be allowed on board if they don't meet a [minimum quality criterion](#). The more data you get about different suppliers, you can start weeding out the suppliers who are not performing well enough to be on your ecosystem. **Governance** of the ecosystem becomes really important.
- And because of all of these things, you start to see **feedback loops** where customers encourage suppliers, suppliers encourage customers. You don't even need to be in the middle of all of this. All of them start interacting with each other and start growing your business.
- And all of this gets amplified even further because your platform is benefiting from **machine learning**. It's capturing information about how customers are making purchases, how suppliers are performing. It takes all of this information and is then able to predictively orchestrate the ecosystem. It is able to make new matches for the first time. You start getting more value because you own all of this information. When you're building your infrastructure, you need to ensure that the core components of machine learning are in place because that is what helps you get sustainable [competitive advantage](#) in a platform business model.

[Pipes VS Platforms | Sangeet Paul Choudary](#) (6 min)

What's good for the platform is bad for the ecosystem

"If we really want to harness this platform revolution well, if you want to become a sustainable platform firm, it's important to not just benefit from the business model but also to figure out a model of governance that works for the benefit of the ecosystem."

1. **Openness** is about how open a platform is. A lot of people don't like closed platforms. Openness today is seen as an ideological debate while it is a business question. When Android started, it started as a completely open platform. What happened was that Amazon and Samsung came in, they forked Android in different directions and Android realized it did not have control over the ecosystem. Android changed this openness to have control over critical services like Maps because without Maps a smartphone is useless and it can control the user data. When platforms talk about openness, they are really looking at ways in

which they can close the platform to control the key critical points of ownership on the platform.

2. Today, many platforms work in ways that are not interoperable. A great example is healthcare. The best way of doing healthcare would be to allow the patient to own the data, make that data interoperable across all platforms and allow the patient to move easily from one platform to another. But **interoperability** comes into the way of a platform's profit. Because the fact that a platform is not interoperable is what creates lock-in for the consumer. Because if my data remains with the hospital, it's in the best benefit of the hospital, because I stay locked in with the hospital. When I go to a new hospital, I cannot take the data with me and that prevents me from moving to a new hospital.
3. A lot of platforms have data that can be used to manipulate the ecosystem. There are ways in which Uber manipulates the market, not in a negative way they claim. They did some research and found out that when someone's mobile phone battery is dying out, people are more likely to accept 9-10 times higher price than those who still have some battery left in their mobile phone. That's a great data point if they want to **manipulate the market**.
4. We think about **trust** between participants on the platform. But more importantly it is also important to understand in the context of how producers on the platform trust the platform. There is [Harvard Business School research](#) that lays out how Amazon looks at data about different merchants selling goods on the platform. It looks for items that sell really well with Amazon's single-day-delivery. Very often what Amazon does is, it goes ahead and acquires these production facilities and starts putting out its own products on the platform which over time edges out these merchants. It is very hard for these merchants to trust the platform because of the power the platform has. Another example comes from Uber who raised their commission rates from 20 to 25% on average (as Uber claims) while [drivers in Bristol](#) see commission rates over 40%.
5. **Platforms should be neutral** and not side with participants, and do not give certain participants too much power. Google is facing lawsuits in the EU because it is not seen as neutral. Very often Google gives higher placement to its own services than to the services of competitors. For the platform to really create value in a market and to ensure the ecosystem gets the most values, it should do it in a way that it stays neutral to all interactions that happen in the ecosystem.
6. At this point in time, most successful platform implementations are funded by venture capital. And all of these platforms are measured on **return on investment**: what have we invested in the platform and how much can we get out of it. And

this is where the big question comes up. Yes, platforms as a business model can create a lot of value but will we find financing schemes and governance mechanisms that will ensure that platforms are actually to solve the world's biggest problems. The question of impact remains unanswered because today, most of the platforms that we see are still not working on that particular model.

[What's good for the platform is bad for the ecosystem - Sangeet Paul Choudary](#) (7 min)

Communities in an ecosystem

In biology, a community is made up of populations of different species or organisms. These are biotic factors such as plants, animals, and bacteria. While an ecosystem is made up of a community, it also includes the abiotic factors in the environment such as the temperature, water, and the landscape of the area.

If an ecosystem is a community that consists of actors playing inside a shared environment, then what's important is to understand how to foster a healthy community. A sense of community can be attributed to four ingredients:

1. **Membership** requires the community to have a well-defined territory. [The territory, or boundary, doesn't have to be physical](#). It can be flexible and change over time as the community evolves. However, it has to represent something familiar to all the members of the community. In the end, the goal of the boundary is to identify who belongs to the community and who doesn't.
2. The community can be considered elastic in the sense that it adjusts to the power of **influence** coming from its members. In turn, each member has to earn the trust of the community as a whole. Without trust, members are not able to exercise any influence and are instead seen as a threat. This game of balance is what generates what is commonly called a sense of community. The more the balance leans toward your ability to influence, the more you feel attracted to the community. On the other hand, the more power to influence you have, the more trust the community needs to have in you, so it's a balancing act. Finding the right spot in the balance between power and trust is not an easy task and can be seen as one of the motivational factors that make members feel attracted to the community.
3. If a symbol is considered a suitable identifier of a community, there are other things that can help members feel that they're considered part of the larger group. [...] Another way of fostering **integration** is by providing a shared community economy where different members can easily exchange resources.

People want to be a part of a community so they can meet their needs, whatever they are. Trading, or exchanging goods and services, is an excellent way to meet people's needs inside a community.

4. Trust and influence help community members reach a state of **shared connection** where members feel that they are all part of a bigger whole. [...] Communities that reach the state of shared connection are much harder to break apart than the ones that don't. [...] However, to be effective, the shared connection has to come from something fundamental to all the participants.
 - a. The first thing to consider is the quality of the interactions between community members. People join and become a part of a community to get something in return, and if interactions are not perceived as valuable, then members leave.
 - b. Another thing that helps reach a state of shared connection is making sure that all the community events—time that members spend together reaching a common goal—have closure. Closure brings trust among the involved participants and fosters a sense of accomplishment and purpose, which makes members feel that they're all sharing something bigger than themselves.
 - c. Finally, making sure that there's an agreed-upon mechanism of honor will help foster the state of shared connection. When members feel that they can be publicly rewarded for their actions, they will have a stronger sense of belonging.

[Difference Between a Community and an Ecosystem](#)

[Platforms are communities](#) (2020)

[Sense of community: A definition and theory](#) (1986)

Community management is the new human resources management

Community management is often viewed as an extension of marketing. If a linear world demanded marketing and customer relationships to manage and influence an audience, popular thinking would have us believe that a networked, participatory world should shift the focus of marketing to community management.

But a community isn't simply a more participative audience. A community must be scaled in a platform business, in much the same way that a workforce of employees is scaled within an organization.

Community management requires structuring and managing incentives for participants, enabling the learning and development of producers, and creating a host of other

support infrastructure that the human resources department would traditionally provide to an organization internally.

Managing community incentives and governance is as important as managing internal employee conduct and compliance.

Example: Facebook focusing on connections

Facebook, for example, discovered that users found the platform valuable only after they had connected to a minimum number of other users. Until then, they were likely to stop using the network entirely. In response, Facebook shifted its marketing efforts away from recruiting new members to helping them form connections.

[Platform Scale, Sangeet Paul Choudary](#) (2015)

DX is the new UX

UX emerged as a key competitive battleground in a world of many user interfaces (UIs) competing for the scarce resource of user attention.

DX is emerging as the next big battleground in a world of many application programming interfaces (APIs) competing for the scarce resource of developer commitment.

Yes, DX has always been important. So, what's new now?

To understand what's different, we need to understand the difference between developer-led and developer-extended platforms. The majority of developer engagement over the past four decades (since the dawn of personal computing) has been on developer-extended platforms.

These platforms follow a common playbook. They launch a core product with superior UX to gain traction on the consumer side. They use consumer traction to attract developers. Developers extend the core product making it more usable (scale) for a broader market (scope) of consumers. Consumer traction increases, and the flywheel sets in.

The iPhone is a great example of the above phenomenon. The iPhone itself was a great product that rapidly built a large customer base. This attracted developers on board who created a host of apps to extend phone functionality, enabling users to customize their phones based on the apps they downloaded.

Developer-led platforms are fundamentally different. Instead of relying on consumer engagement to attract developer commitment, they rely on [developer commitment](#) to

attract consumer engagement. Their playbook looks something like the following: They launch a core toolset with superior DX to gain traction on the developer side. They use developer traction to build value for consumers. Consumer engagement increases as developers build new functionality (scale and scope) around the core toolset. Developer traction increases as market adoption of the platform among consumers grows, and the flywheel sets in.

Developer commitment is the key source of competitive advantage for developer-led platforms. The core toolset doesn't get consumer traction. Developers come on board and extend it towards consumer usage. In this way, developers don't simply deepen an already existing moat for the business. They help dig the moat in the first place.

DX is the new UX

Example: SAP community incentives & roadmap for developers

The enterprise management platform company SAP uses a social currency like that of iStockphoto or Stack Overflow to motivate developers to answer one another's questions. Points earned when the employee of a development company answers a question are credited to a company account; when the account reaches a specified level, SAP makes a generous contribution to a charity of the company's choice. The system has saved SAP \$6–8 million in tech support costs, generated numerous new product and service ideas, and reduced average response time to thirty minutes from the one business day that SAP promises. SAP estimates that knowledge spillovers from these activities account for a half-million-dollar gain in annual productivity for a typical enterprise software partner. Even more interesting, SAP has used the social currency supply to stimulate its developer economy in the same way as the Federal Reserve uses the money supply to stimulate the U.S. economy.

When SAP introduced a new customer relationship management (CRM) product, it offered double points on any answer, code, or white paper relating to CRM. During the two-month duration of this "monetary expansion" policy, developers found gaps in the software and devised new features at a vastly higher rate. Used as a money supply, the increased flow of social currency caused overall economic output to rise. In effect, SAP employed an expansionary monetary policy to stimulate growth—and it worked. In addition to promoting economic growth, well-designed market mechanisms can incentivize the creation and sharing of intellectual property and reduce the riskiness of interactions on the platform. Beautiful, useful ideas are public goods. Which raises the question: what is the optimal intellectual property policy for a platform business? If a developer working on a platform invents a valuable idea, who should own it, the

developer or the platform? It's possible to imagine arguments on both sides of the issue. Giving ownership to the developer provides incentives for idea creation. Giving ownership to the platform facilitates standardization and sharing and enriches the platform ecosystem as a whole.

State-mandated laws regarding patents and other forms of intellectual property protection are clumsy and expensive to enforce. A more elegant platform-based solution is required. SAP has tackled this problem through two practices. First, it publishes an 18–24-month advance road map indicating what new products and services it intends to build to enhance its offerings to its corporate clients. This not only tells SAP's outside developers what digital real estate will be available for their own innovations but also gives them up to two years before they face competition. The two-year window thus serves as a metaphorical patent period. Second, SAP has made a policy of partnering with developers financially or buying them out at a fair price. This assures developers that they will be fairly compensated for their work, [reduces partner risk, and encourages outside investment in the SAP platform](#).

Marketing & platform ecosystems

In 2013, Google and Apple (platform businesses) passed Coca Cola (pipeline business) as the [most valuable brand](#). Google spends less on advertising globally than Coca Cola spends alone in the US. The reason appears to be interactivity. You touch the product and interact with it so commonly. Kids know what it means to google. The interactivity means they don't have to brand; they don't have to spend on it.

Platforms invert the firm. Because the bulk of a platform's value is created by its community of users, the platform business must shift its focus from internal activities to external activities. In the process, the firm inverts—it turns inside out, with functions from marketing to information technology to operations to strategy all increasingly centering on people, resources, and functions that exist outside the business, complementing or replacing those that exist inside a traditional business. The language used to describe this process of inversion differs from one business function to another. In marketing, for example, Rob Cain, the CIO of Coca-Cola, notes that the key terms used to define systems of message delivery have shifted from broadcast to segmentation, and then to virality and social influence; from push to pull; and from outbound to inbound. All these terminological changes reflect the fact that marketing messages once disseminated by company employees and agents now spread via

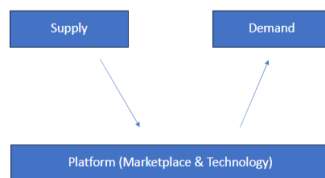
consumers themselves—a reflection of the inverted nature of communication in a world dominated by platforms.

[Marshall Van Alstyne- How Network Business Models are Changing the Shape of Industry](#) (min 4:00)

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Marketing a platform

As a quick reminder, in the platform economy, there are (at least) 2 target groups: suppliers & users.



“Platform marketing is used to drive awareness, trust and interest, but it also gives you an opportunity to get product management feedback about your platform. That last part is one of the underappreciated parts of advocacy.”

[Platform marketing, advocacy, DevEx, documentation, and Backstage - Tanzu Talk](#) (56 min)

Platform Marketing = Reputation Marketing?

In the fall of 2014, the New York City subways were suddenly filled with ads for a business many city residents were just beginning to learn about—Airbnb. These were not ordinary ads aimed at convincing potential customers to try out the room rental services provided by Airbnb. They were what advertising people call corporate image ads, designed to burnish the reputation of the company itself. “Airbnb is great for New York City,” read the slogan on every ad.

Sidenote: That didn’t go well because [not everything that is good for the platform is good for the ecosystem](#).

But not all subway riders agreed. Within days, many of the ads had been “edited” by marker wielding graffiti writers with their own thoughts about Airbnb. Journalist Jessica Pressler recorded some of the choicest comments in New York magazine. One poster had been supplemented with the observation, “Airbnb accepts NO Liability.” Another had been marked with the scrawl, “The dumbest person in your building is passing out a set of keys to your front door!” And on several posters, the phrase “for New York City” had been replaced with a different handwritten conclusion: “Airbnb is great for Airbnb.”

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

What is the product?

A business that goes about building a platform the way it would build a pipe is setting itself up for failure. Many business leaders erroneously apply the pipe execution model to building platforms. The media industry is struggling to come to terms with the fact that the model has shifted. Traditional retail, a pipe by most measures, is being disrupted by the rise of marketplaces and in-store technology.

Platforms require completely different mental models to succeed. **They need interaction-first thinking.** Pipes rely on user-first thinking, not interaction-first thinking. In user-first thinking, the single user's perspective rules all business decisions. This perspective works well when value flows linearly from the business to its users. As businesses move toward enabling interactions between producers and consumers of value, they must adopt interaction-first thinking. In interaction-first thinking, the focus on users does not cease but becomes subservient to the focus on interactions.

In an age of platform hype, everyone scrambles to find the platform in their product. If you're a product business, chances are you've been told that platforms beat products and hence you should look to become a platform. There is a problem with over-emphasizing platform thinking and the 'platforms beat products' narrative: By positioning 'platforms' as superior to 'products', we risk throwing away the simple first principles of product thinking which have served us well for centuries. We need to embrace 'product thinking' and find the product in our platform. Understanding the 'product' in our platform helps us clarify where it really creates value, and more importantly, where it can effectively capture value.

A product is the offering the firm creates through the transformation of inputs, which it then distributes downstream. This is fairly straightforward in product value chains. It gets a lot more vague with platforms. Platforms combine:

- a mechanism to attract and organize inputs (on the producer side) and
- a mechanism to distribute (on the consumer side).
- But what is the 'product' in the middle in case of platforms?

[Finding the product in your platform](#)

[Platform Scale, Sangeet Paul Choudary](#) (2015)

Example: Spotify

What does Spotify really 'produce'? You might say that Spotify's product is the music streaming service. But that's really just a proprietary distribution technology in a product value chain. You might argue that the products are the songs. But that's not what

Spotify is really producing. The labels own the songs, the artists create them. The songs are inputs into the ‘production’ process. What exactly is the ‘product’ in Spotify’s platform? **Spotify’s ‘product’ is the playlist.**

Playlists are central to Spotify’s strategy within the music ecosystem. Songs, which were traditionally bundled into albums, were unbundled by the internet and then went through three phases of free distribution through file-sharing services like Kazaa and Napster, pay-per-tune distribution through iTunes, and fixed-fee unlimited catalog distribution through Spotify. This unbundled song distribution changed for the first time with Spotify’s playlists. Playlists provide the locus of rebundling where songs – unbundled from albums and artists – are rebundled into theme-based playlists.

Playlists are the core product driving discovery of new artists. They are critical for Spotify to gain a right to sit in the middle of the music value chain and not be merely a digital distributor for the top 3 music labels.

What Spotify ‘sells’ to advertisers and brands is influence and targeting through playlists.

1. The “Sponsored Playlist” program allows brands to sponsor a Spotify-curated playlist for 1 week.
2. Using the metadata of playlists, Spotify also launched “Branded Moments,” a program which leverages the real-time insights playlists offer into activities that listeners are engaged in. E.g. a fitness brand targeting a playlist labeled Workout.

The playlist – the central locus of rebundling in the music streaming business – is Spotify’s core product.

Spotify takes songs/tracks as ‘inputs’, rebundles them into playlists as ‘product’, and sends them down its proprietary streaming ‘distribution’ channel. And all of this helps Spotify gain power back from the labels. Without the playlist, Spotify is merely a glorified distributor in the music value chain. A scaled out one, no doubt, but at the end of it all, just a glorified distributor dependent on the whims and fancies of a highly concentrated supplier base constituting three music labels.

The playlist is the ‘product’ that gives Spotify the right to be a platform.

[Finding the product in your platform](#)

Growth Marketing

In its simplest form, growth marketing, also known as growth hacking, is marketing that is hyper-focused on, well, growth. Its very existence is a response to rapidly growing technology, a shrinking world, and a culture that seems to change on a daily basis. Growth marketer Sean Ellis says a “growth hacker’s true compass is north.” Bigger, better, and more are what it’s all about.

Growth marketing is anything but old school. It differs from traditional marketing in that it uses modern tools and metrics to measure the performance of specific strategies and tactics as completely as possible. Different from old-school marketing, growth marketing is inherently tailor-fit to your company, and constantly evolves to fit changing customer attitudes, respond to industry events, and handle your competition. It’s completely dynamic. In the past, marketers used generally accepted principles to guide their efforts, and worked creatively within those parameters. Growth marketing, however, depends on thinking outside the box. It’s highly experimental. It’s where marketing meets scientific theory. Growth marketers don’t build a marketing strategy to implement, because their strategy is to adapt to whatever seems to be working best.

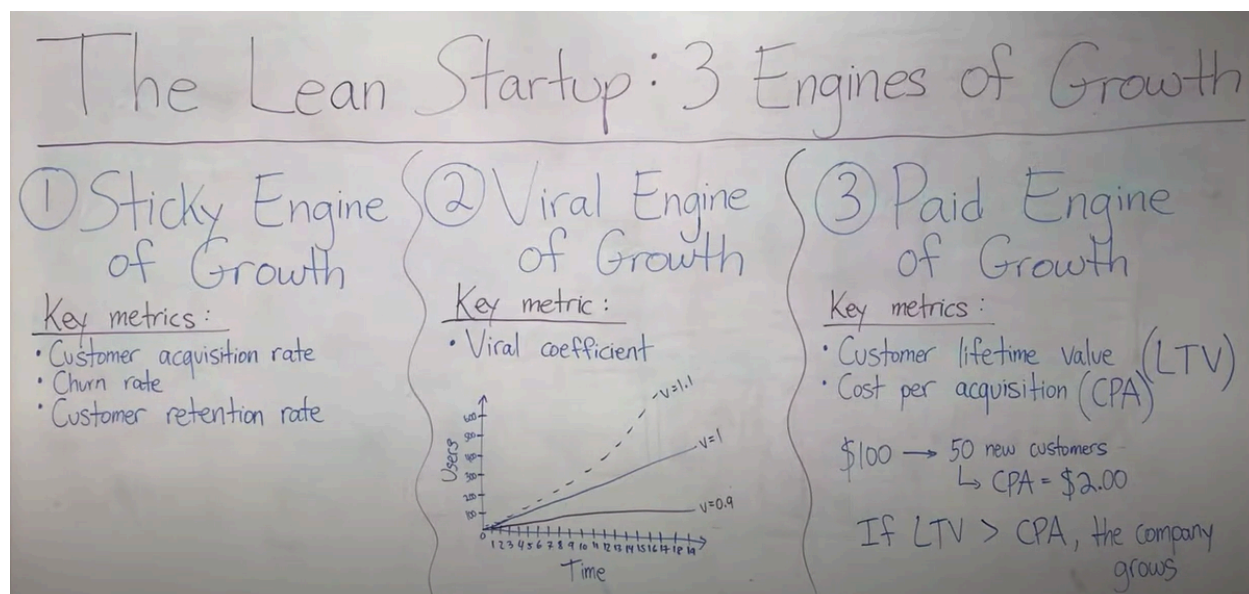
Key aspects of growth marketing include:

1. Data-Driven Decision-Making
2. Experimentation and Testing
3. Customer-Centricity
4. Agile and Iterative Approach
5. Holistic Funnel Optimization (rather than focusing solely on customer acquisition)

[What is Growth Marketing? Strategies to grow your business](#)

3 Engines of Growth

An engine of growth is the mechanism that startups use to achieve sustainable growth. Each engine described has an intrinsic set of metrics that determine how fast a company can grow.



1. **Sticky Engine of Growth:** Stickiness is measured as attrition rate or churn rate. The churn rate is defined as the fraction of customers in any period who failed to remain engaged with the company's product. If the rate of new customer acquisition exceeds the churn rate, then the company will grow. A startup must focus on customer retention as much as possible.
2. **Viral Engine of Growth** is powered by a feedback loop which measured by a viral coefficient which is a numerical measure of how much virality a product or a company has. The higher the viral coefficient the faster the product will spread. The viral coefficient measures how many new customers will use a product as a consequence of each new customer of each new customer who signs up. Example: Hotmail built in a marketing feature in their product. At the bottom of every single email that a user sent out they added the sentence "PS Get your free email at Hotmail along with a clickable link". 12 months after launching the service they already had 12 million subscribers.
3. **Paid Engine of Growth** is another type of feedback loop. Each customer pays a certain amount of money over his or her lifetime as a customer. Once variable costs are deducted this is usually called the customer lifetime value (LTV). That revenue can be reinvested into the company for growth, for example to buy more advertising. The margin between the LTV and the CPA (cost per acquisition) determines how fast the paid engine of growth will turn.

Scaling network effects

It's important to distinguish network effects from other familiar market-building tools, such as **price effects** and **brand effects**. Misunderstanding of these distinctions is a source of the current confusion over how to value platform business models and contributed to the dot-com boom and bust of 1997–2000.

The problem is that **price effects** are evanescent. They disappear the moment the discounts end or another firm offers a better price. Typically, only 1–2 percent of customers convert from free to paying. Thus, as David Cohen, CEO and founder of the venture incubator Techstars, says, you need to reach millions of customers before the giveaway model becomes profitable. Freemium models also create freeloaders than can be hard to monetize (that is, to profit from), as FreePC discovered in 1999 when it gave away free Pentium PCs in exchange for viewing ads and the prospect of online sales.

Brand effects are stickier. They arise when people come to associate a particular brand with quality. But brand effects, like price effects, are often difficult to sustain. They can also be extremely expensive. EToys spent millions to establish a brand in hopes of competing with Amazon and Toys“R”Us. Kozmo, an online company that promised free one-hour delivery of food, books, coffee, and other basic goods in major U.S. cities, hired actress Whoopi Goldberg as a spokesperson and paid her in stock, only to have the business collapse soon thereafter. In January 2000—the peak before the dot-com crash—nineteen startups bought Super Bowl ads, spending over \$2 million each in order to build brand recognition. A decade or so later, eight of them no longer existed.

Price effects and brand effects have their place in a startup's growth strategy. But only [network effects](#) create the virtuous cycle we described above, which leads to the building of a longlasting network of users—a phenomenon we called lock-in.

Another growth-building tool that's easy to confuse with network effects is **virality**. Derived from the expression “going viral,” virality is the tendency of an idea or brand to be circulated rapidly and widely from one Internet user to another.

Network effects depend on the size of the network. So, one important corollary is that effective platforms are able to expand in size quickly and easily, thereby scaling the value that derives from network effects. It's hard to remember now, but there was a time when Yahoo was a more popular portal to the Internet than Google.

As the story of Google suggests, networks that permit frictionless entry are able to grow organically almost without bound. Frictionless entry is the ability of users to quickly and

easily join a platform and begin participating in the value creation that the platform facilitates. Frictionless entry is a key factor in enabling a platform to grow rapidly.

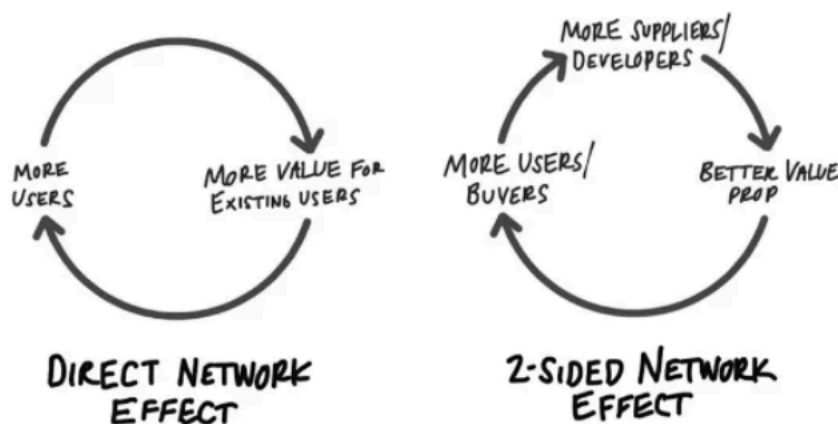
Scaling a network requires that both sides of the market grow proportionally. For example, one Uber driver can serve an average of about three Uber riders an hour. It would make no sense for Uber to have one rider and 1,000 drivers—nor 1,000 riders and one driver. Airbnb faces a parallel issue in scaling both hosts and guests. If one side becomes disproportionately large, coupons or discounting to attract more participants to the other side becomes good business.

In some cases, the growth of a platform can be facilitated by an effect we call side switching. This occurs when users of one side of the platform join the opposite side—for example, when those who consume goods or services begin to produce goods and services for others to consume.

On some platforms, users engage in side switching easily and repeatedly. Uber, for example, recruits new drivers from among its rider pool, just as Airbnb recruits new hosts from among its guest pool. A scalable business model, frictionless entry, and side switching all serve to lubricate network effects.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Types of network effects



A two-sided network (i.e., one with both producers and consumers) has four kinds of network effects. It's important to understand and consider all four when designing and

managing a platform. In a two-sided market, **same-side effects** (direct network in the screenshot above) are network effects created by the impact of users from one side of the market on other users from the same side of the market—the effects that consumers have on other consumers and those that producers have on other producers. By contrast, **cross-side effects** are network effects created by the impact of users from one side of the market on users from the other side of the market—the effects that consumers have on producers and those that producers have on consumers. **Both same-side effects and cross-side effects can be positive or negative**, depending on the design of the system and the rules put in place.

[Growth Engines in Marketplaces & Platforms](#)

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Example: Instagram grew on external networks

Everyone has heard of Instagram's rapid growth—over 100 million active users in less than two years—which led to its billion-dollar acquisition in April 2012 by Facebook. What is less well known is that Instagram achieved this rapid growth without employing a single traditional marketing manager. It happened because the company's platform was carefully designed to make viral growth organic and practically inevitable. Unlike its competitor Hipstamatic, Instagram didn't simply allow users to save, organize, and filter pictures. It encouraged them to share their photos on external networks like Facebook, converting a single-user activity into a social, multi-user activity. Every time users engaged the app, they shared their creations. Every point of app usage became an instance of app marketing. In essence, Instagram converted all its users into marketers.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Example: Paypal – using an existing community to grow

They (= founders Levchin and Thiel) realized they would have to overcome one of the toughest challenges associated with creating a business designed to serve two sides of a market—the chicken-or-egg problem. [...] In a lecture he later gave at Stanford, Peter Thiel explained what happened next: PayPal's big challenge was to get new customers. They tried advertising. It was too expensive. They tried BD [business development] deals with big banks. Bureaucratic hilarity ensued. ... the PayPal team reached an important conclusion: BD didn't work. They needed organic, viral growth. They needed to give people money. So that's what they did. **New customers got \$10 for signing up, and existing ones got \$10 for referrals. Growth went exponential, and PayPal wound up paying \$20 for each new customer. It felt like things were working and not working at the**

same time; 7 to 10 percent daily growth and 100 million users was good. No revenues and an exponentially growing cost structure were not. Things felt a little unstable. [...] in the end, the strategy worked. PayPal dramatically increased its base of consumers by incentivizing new sign-ups. Most important, the PayPal team realized that getting users to sign up wasn't enough; they needed them to try the payment service, recognize its value to them, and become regular users. In other words, **user commitment was more important than user acquisition**. So, PayPal designed the incentives to tip new customers into the ranks of active users. [...] Once users experienced the convenience of PayPal, they often insisted on paying by this method when shopping online, thereby encouraging sellers to sign up. **New users spread the word further, recommending PayPal to their friends. Sellers, in turn, began displaying PayPal logos on their product pages** to inform buyers that they were prepared to honor this method of online payment. [...] However, the company leaders didn't sit back and rely on the positive feedback loops to do all the growth work on their own. They looked for opportunities to jack up the growth rate still further. In early 2000, they noted the growing popularity of PayPal on eBay, the most popular online auction site. It was a natural place for PayPal, since most of the sellers on eBay aren't full-time merchants but ordinary people without facilities for accepting credit cards or other forms of online payment. PayPal's marketing team (see [*Paypal's framework for segmentation in complex B2B markets*](#)) opportunistically refocused its efforts toward enabling payments on eBay. Among other techniques, they simulated consumer demand on eBay by creating a bot (an automated software tool) that bought goods on the site and then insisted on paying for these transactions using PayPal. Noting this apparent growth in demand, many eBay sellers signed up for the PayPal service—which in turn made PayPal even more visible and attractive to consumers. The sellers began posting PayPal icons on their sites, enabling buyers to access the payment system with just a single click of the mouse, reducing friction still further. Within three months, PayPal's user base grew from 100,000 to one million.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

[Confinity which competed with Elon Musk's X.com](#)

Negative network effects

The growth of a network can produce negative network effects that drive away participants, even leading to the death of a platform business. One negative network effect occurs when the growth in numbers that enables more matches between producers and consumers also leads to increasing difficulty, or impossibility, in finding the best match. To avoid this dilemma, **frictionless entry must be balanced through effective curation**. This is the process by which a platform filters, controls, and limits the

access of users to the platform, the activities they participate in, and the connections they form with other users. When the quality of a platform is effectively curated, users find it easy to make matches that produce significant value for them; when curation is nonexistent or poorly handled, users find it difficult to identify potentially valuable matches amid a flood of worthless matches.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Example: OKCupid – using matching algorithms

To solve this problem, OkCupid (*dating platform*) implemented a curation strategy involving multiple levels of network matching. The first level addresses the obvious issue of matching compatible interests. Do both parties smoke? Do both parties like tattoos and horror movies? Do both parties believe in dinosaurs? This level eliminates many clearly unsuitable matches and reduces the number of participants in the process.

The second matching level addresses the question of comparative attractiveness—the “in her league” question. If OkCupid’s algorithm determines, based on reactions by other users, that Joe is significantly less attractive than Mary (for example), then Joe’s routine search for matches will not turn up Mary’s picture. (She might show up in a highly targeted search, but not otherwise.)

Instead, Joe will be presented with a selection of women thought to be comparable to him in attractiveness. The result is win-win. Mary is happier because the platform helps her find what she’s looking for while protecting her from an onslaught. And Joe is happier, too, because women are returning his messages when previously he got the cold shoulder.

Of course, the use of this algorithm means that, if a guy only sees pictures of average-looking women when conducting an OkCupid search, he probably doesn’t have the movie-star good looks he thought he had. But his chances for a successful match have greatly increased, which should lead in the long run to a higher level of satisfaction.

[Platform Revolution](#): Geoffrey G. Parker, Sangeet Paul Choudary, Marshall Van Alstyne (2016)

Customer Education Marketing

“A webpage dedicated solely to the features of a product will not be successful on its own. Customers want context; they want to know about the product as it exists in the real world. They want to know which problems that product will solve, and what experts

in the field think of its merits. What we're talking about is insight. And in a world gone mad with ever-more-elaborate marketing ploys, it's more important than ever before."

"Companies today can no longer rely on a small percentage of early adopters to create a market and future success by opening the minds of the next cohort of customers. With short, competitive windows and technology cycles, a high percentage of customers must embrace the technology quickly or the innovation will languish and soon be forgotten and disappear. Customer behaviours should change in concert with product, service or process innovation. Thus, companies need to master managing customers' behaviours if the uptake of an innovation is to be accelerated and revenue objectives met."

"In any business, building trust with customers is essential for long-term success. When customers trust a business, they are more likely to make purchases, recommend the business to others, and remain loyal over time. One effective way to build trust with customers is through customer education. When we talk about customer education in the context of growth marketing, we are referring to the process of providing customers with information and resources that help them understand a product or service and how it can benefit them. This involves more than just marketing and advertising - it's about helping customers make informed decisions and empowering them to get the most out of their purchases."

[The importance of customer education in growth marketing](#)

[Don't Market To Your Customers: Educate Them Instead](#)

[The marketer's challenge: how to teach customers new behaviours](#)

Data usage in Marketing

Information consumption preferences

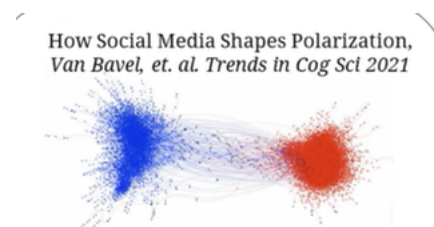
*What information consumes is rather obvious: it consumes the attention of its recipients. Hence **a wealth of information creates a poverty of attention** and a need to allocate that attention efficiently among the overabundance of information sources that might consume it.*

– Herbert Simon 1971

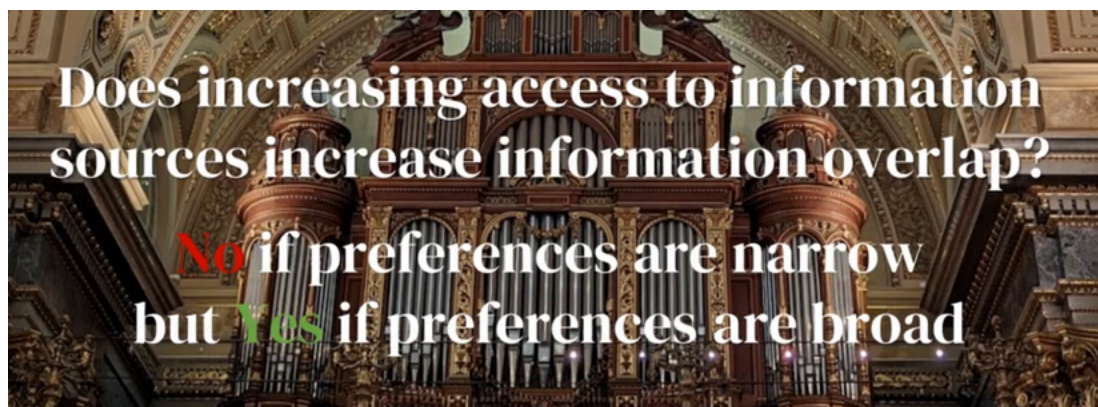
"If we introduce that bounded rationality constraint: Suppose you have only 2 ears. You can only carry at most 2 conversations at a time. You can have a couple of channels. If

we introduce that and then we introduce increasing access. Do we get more overlap? Well, if you increase the access from local to global, if there are preferences for like-minded contact. In fact, the information might decline. There are interesting recent developments in the descriptions in recent literature: the truth decay when people are so overloaded with information, they can no longer separate fact from fiction and fall back on opinion and tribal beliefs in order to make decisions.

Research conducted in 2021 suggest that content platforms lead to polarization and cyberbalkanization.”



“Using Herbert Simon’s work, you can actually see how polarization might have happened in the context of increasing information and yet bounded rationality. The interesting news is that it is not exclusively driven by bounded rationality constraints. It’s actually driven by preferences. If preferences are narrow, if your preferences are for affinity groups, then yes, we do get polarization. But if there are preferences for diversity, then in fact, we get integration.”



Now, what could this mean for marketing and content creation and content distribution using 3rd party content platforms, especially when trying to change a target group’s viewpoint ([silo process to data-driven process](#))?

[Marshall Van Alstyne: Platform Businesses: Past Perspective and Future Prospects - Simon Award 2021](#)

User journeys are the new sales funnels

In a linear world, customers are led through sales funnels. Frameworks representing the customer purchase path – such as the [AARRR](#) (acquisition-activation-retention-revenue-referral) framework that tracks usage metrics – are often designed as funnels.

In a networked world, purchase paths are no longer linear. Instead, users interact with a business across multiple experiences and channels before making a purchase. Even industries like retail, which have lived by the funnel and tracked footfalls religiously, are moving to measure engagement across multiple touch points. In a world of multi-device, multi-channel journeys, the browse and buy experiences are decoupled. It is important to ensure that actions taken by users at various points on this journey are leveraged to personalize their experience at every other point.

Businesses must invest in integrating these user touch points together. A platform serves as an integration layer that connects multiple touch points with the user. Every business that wants to benefit from multi-channel integration and serve the user across this journey must integrate its touch points with users using a platform.

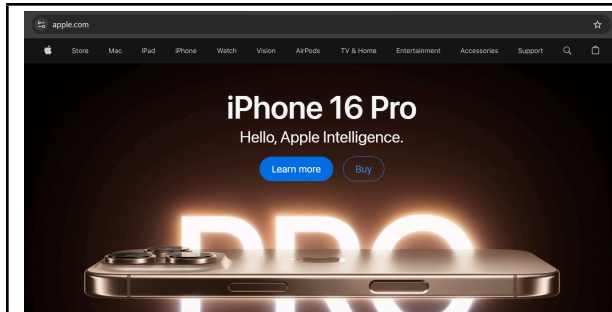
Once integrated, there should be a continuous flow of data across these touchpoints to deliver highly personalized experiences. The platform acts as a sink that constantly absorbs data from the user and consequently delivers highly personalized experiences. Connectivity by data serves as the binding agent between the users' immediate experience and their journey with the business.

[Platform Scale, Sangeet Paul Choudary](#) (2015)

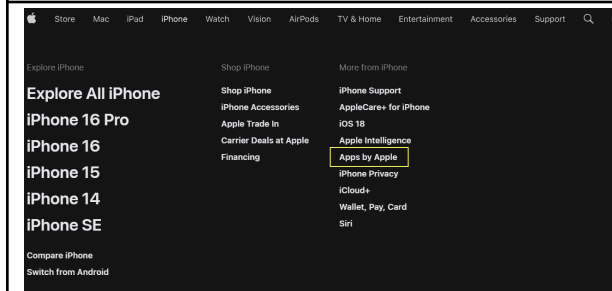
Website examples of businesses with a pipeline/platform overlap

Apple

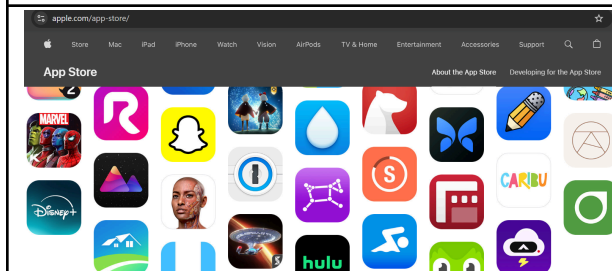
It seems Apple separates software (marketplace) from hardware (ecommerce). Apple stores are run by authorized resellers.



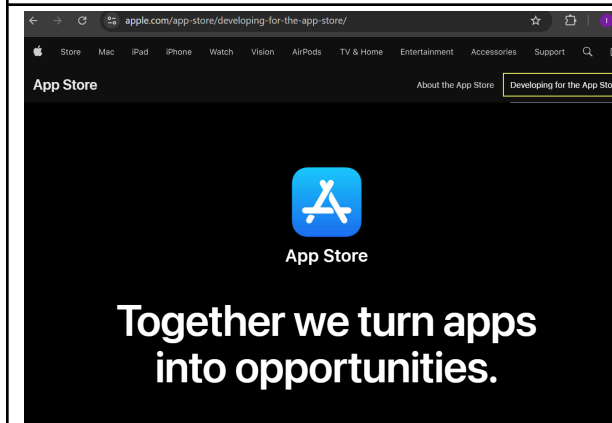
<https://www.apple.com/>



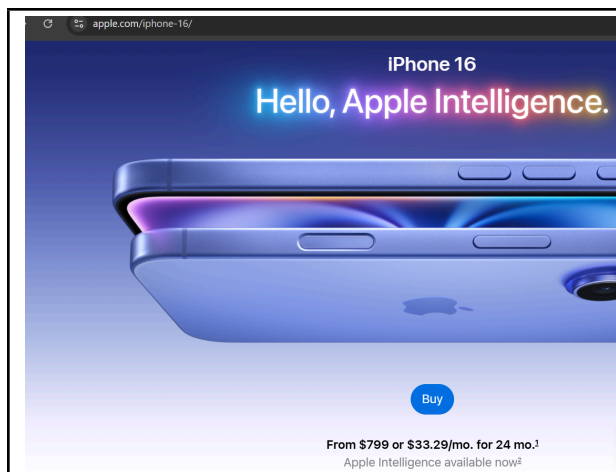
<https://www.apple.com/apps/>



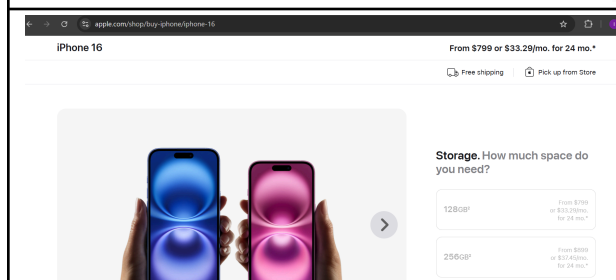
<https://www.apple.com/app-store/>



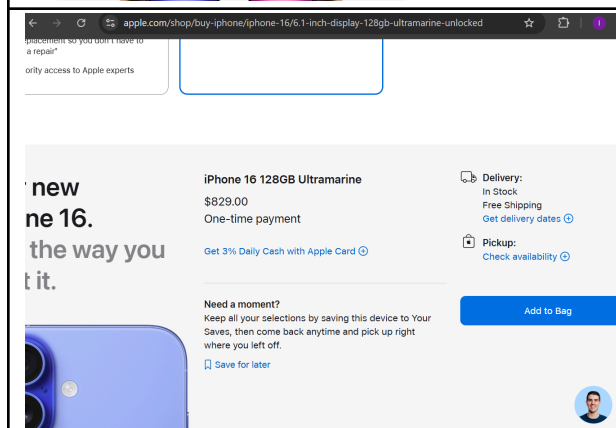
<https://www.apple.com/app-store/developing-for-the-app-store/>



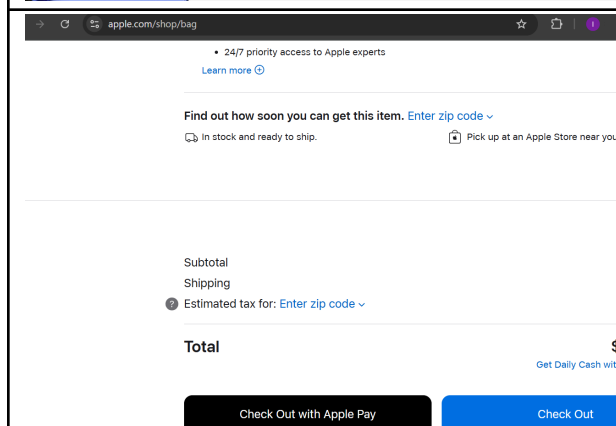
<https://www.apple.com/iphone-16/>



<https://www.apple.com/shop/buy-iphone/iphone-16>



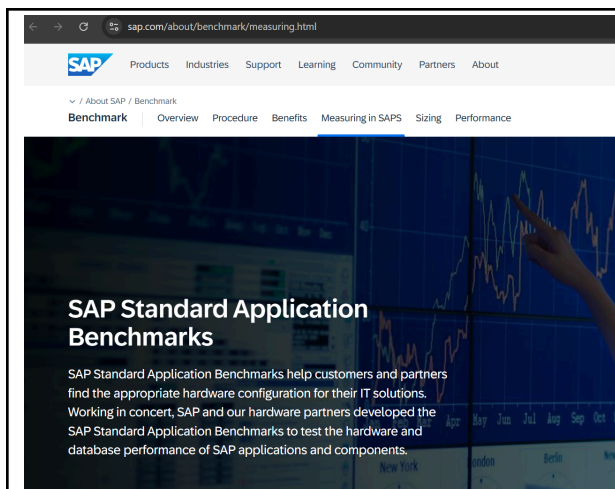
<https://www.apple.com/shop/buy-iphone/iphone-16/6.1-inch-display-128gb-ultramarine-unlocked>



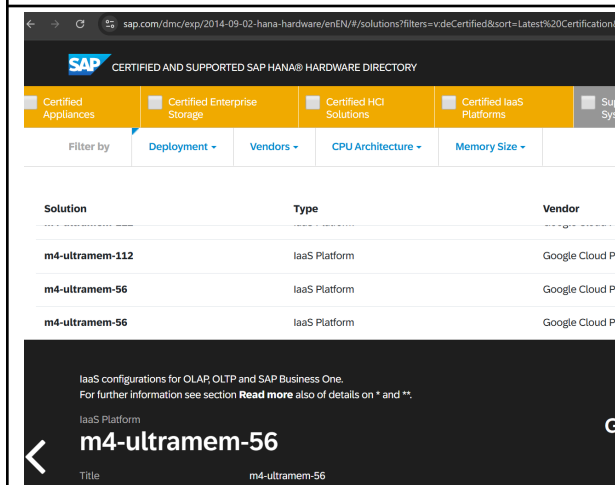
<https://www.apple.com/shop/bag>

SAP

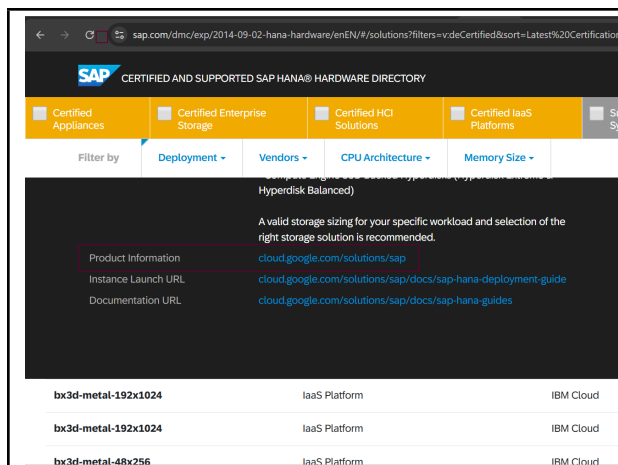
As seen in the [B2B commerce research](#), [SAP's store](#) is their marketplace. They are also applying “match making” (like [OKCupid](#)) to make sure the customer finds the right supplier. Even though SAP doesn't sell hardware, they have ecosystem partners who do sell hardware to run SAP software.



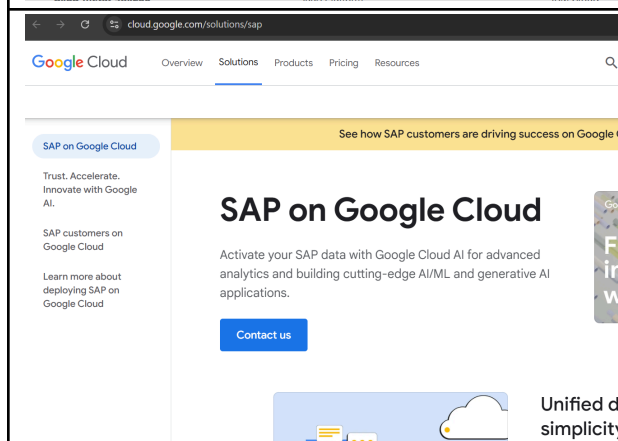
<https://www.sap.com/about/benchmark/measuring.html>



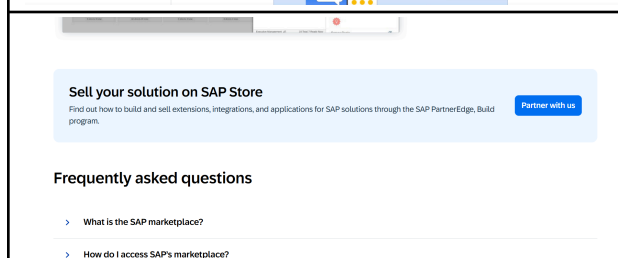
<https://www.sap.com/dmc/exp/2014-09-02-hana-hardware/enEN/#/solutions?filters=v:deCertified&sort=Latest%20Certification&sortDesc=true&id=s:3024>



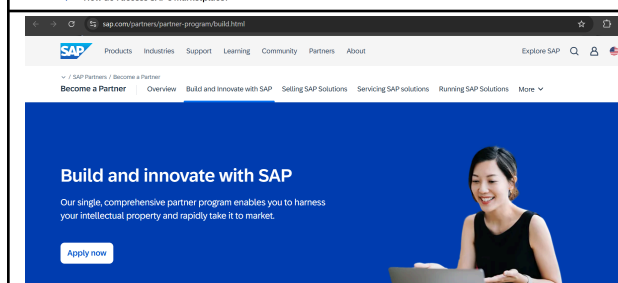
However, the hardware can't be purchased through the SAP marketplace.



But points to the vendor site:
<https://cloud.google.com/solutions/sap>

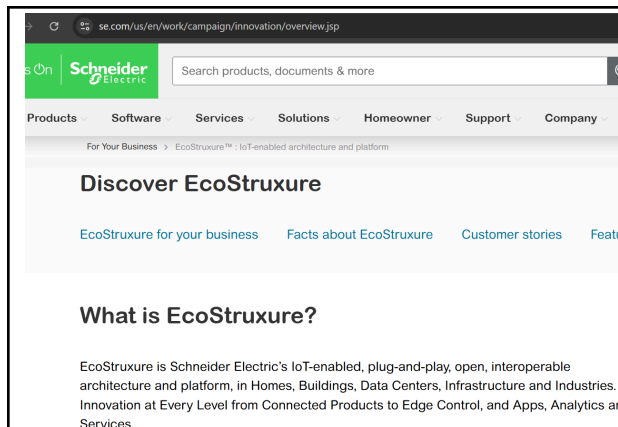


There's a separate path for partners on the SAP marketplace:
<https://www.sap.com/store.html>

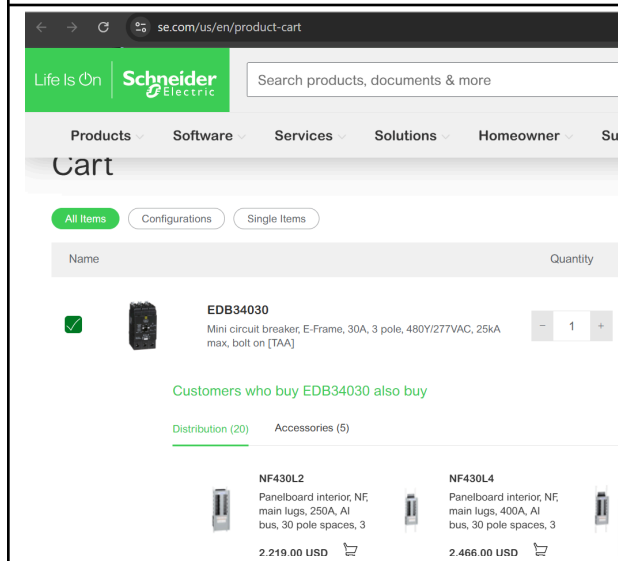


That addresses the different types of partners and requires an application:
<https://www.sap.com/partners/partner-program/build.html>

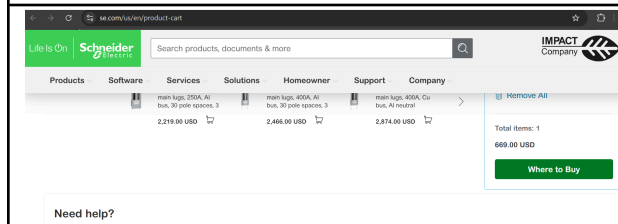
Schneider Electric EcoStruxure



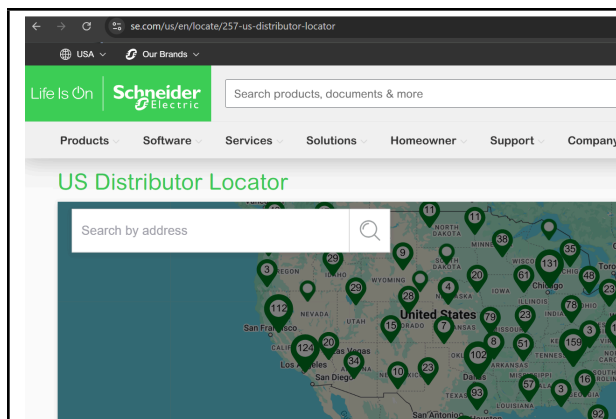
<https://www.se.com/us/en/work/campaign/innovation/overview.jsp>
<https://www.se.com/us/en/work/campaign/innovation/platform.jsp>



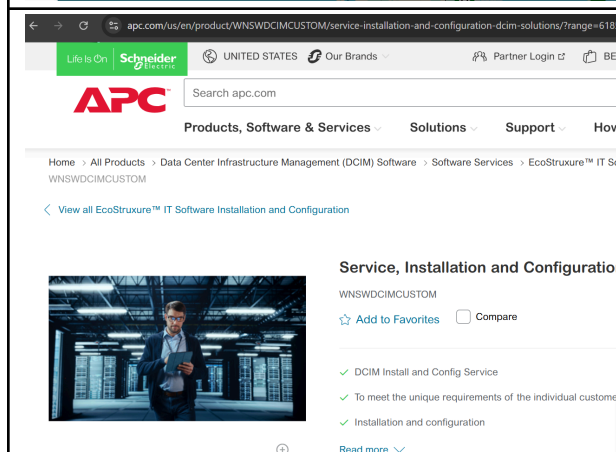
Hardware can be selected, and they use a recommendation engine.



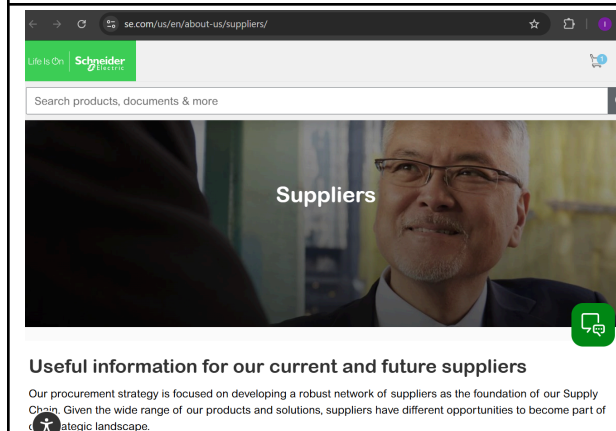
but not all of it can be purchased online



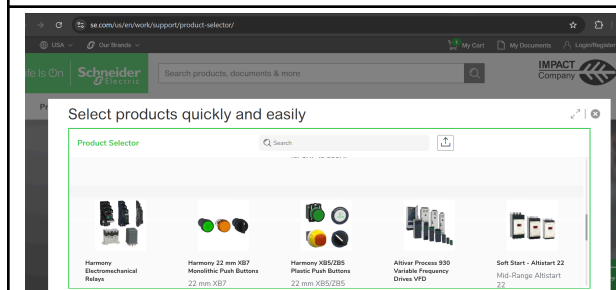
a map of resellers is offered to choose from



Software service packages have their own “product” pages.



They call their partners suppliers
<https://www.se.com/us/en/about-us/suppliers/>



They have a product selector:
<https://www.se.com/us/en/work/support/product-selector/>

VUCA challenge

VUCA is an acronym that stands for volatility, uncertainty, complexity and ambiguity -- qualities that make a situation or condition difficult to analyze, respond to or plan for. Understanding how to mitigate these qualities can greatly improve the strategic abilities of a leader and lead to better outcomes.

The acronym VUCA was first used in the U.S. Army War College in 1987 and was publicly published in 1991 by Herbert Barber. The method was developed after the concepts presented by Warren Bennis and Burt Nanus in their book *Leaders: The Strategies for Taking Charge*. VUCA was applied to the conditions following the end of the Cold War and the conflict in Afghanistan at that time.

The term VUCA can be applied to a situation in general. Sometimes it is used dismissively to discount the value of planning. Someone might unhelpfully say, "Because of the current VUCA, any plans we make will be quickly out of date and useless, so why bother planning?" In this sense, its use is similar to FUD (fear, uncertainty and doubt).

The proper use of VUCA is to apply it to a situation to help quantify risks and create mitigation strategies. Use the VUCA method to go through what is known and not known about a situation or plan. This helps create a better understanding of the situation and what the vulnerabilities and risks are. Using VUCA can help leaders manage the ever-changing modern business landscape. [Tech Target](#)

VUCA solution

As an answer to the VUCA challenge, Vision, Understanding, Clarity, Agility are suggested to overcome the challenges.

- Volatility can be countered with Vision. In turbulence times, shaping a vision is even more important than in normal times. To counter the turbulence while keeping the organization's vision in mind and pass through the economic downturns, or to face new competition in the markets.
- Uncertainty can be countered with Understanding. To be effective in a VUCA environment, leaders must learn to look and listen beyond their functional area of expertise, to make sense of the volatility and to lead with vision. This requires leaders to communicate with all levels of employees in the organization, and to develop and demonstrate collaboration skills.
- Complexity can be countered with Clarity. In a VUCA environment, chaos comes rapidly and hard. There is a sense of urgency to clarify which strategic

orientations you want to explore and focus to get out of the chaotic situation as quickly as possible, keeping consistency with your vision.

- Ambiguity can be countered with Agility. Agility is the ability to communicate efficiently across your organization, decide and move quickly to apply solutions (source : P. Kinsinger & Walch – 2012).
 - Simplification: always try to simplify the way you operate, e.g. reducing the number of interfaces.
 - Keep the flow moving: don't work faster, but better together by reducing white space in your processes on the critical path.
 - Collaborate better together: be more autonomous.

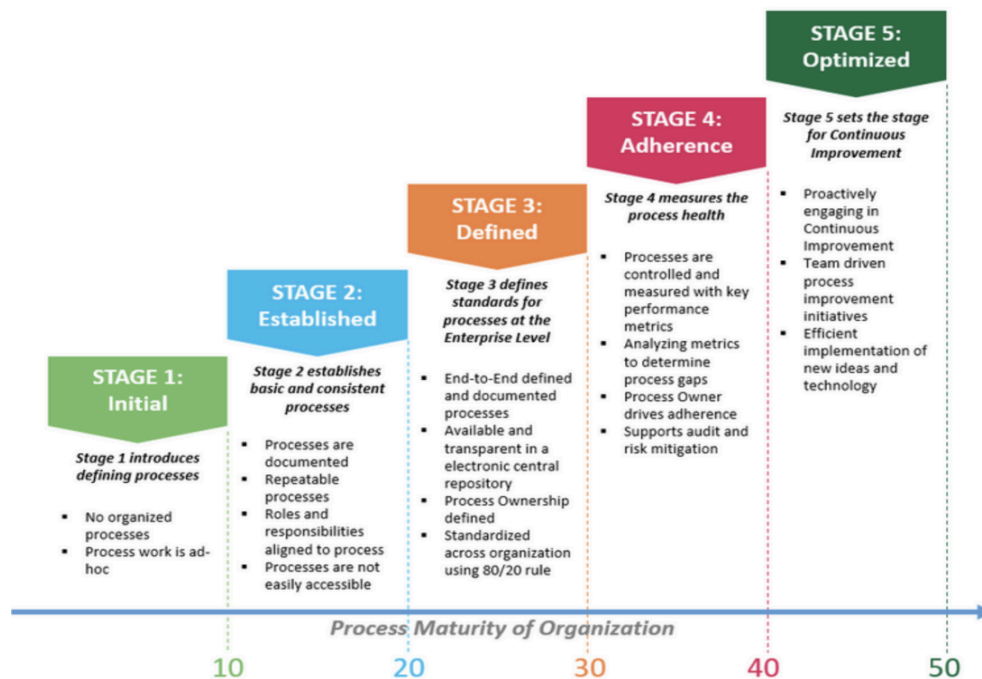
[QRM \(Quick Response Manufacturing\) Institute](#)

[University of Missouri](#) (also suggesting Design Thinking as an approach)

Process management

Many times, organizations implement process management and then ask: “Now what?” They are not sure what happens after the processes are up and running. Other organizations simply assume process-related work is done once the program is operational. Yet, mature organizations understand that the challenge has only just begun when process management is in place; they realize that that the focus must turn to gauging whether process is working and yielding the desired results. Further, they realize that processes can be optimized to yield even better outcomes. Although this can be a difficult challenge—knowing if and how process management is working—using a maturity model can give organizations direction for managing and improving processes as well as answer the question, “Now what?”

[APQC \(American Productivity & Quality Center\)](#)



UC Berkeley

Integrating Design Thinking, Lean Six Sigma, PDCA and Agile

Your Best Bets for Creative Problem Solving - Lean, Six Sigma, Agile, Design Thinking



<https://youtu.be/pr9Mt62okqw?si=vejs-EcpktzEQspi> (15 min)

Since design thinking is a non-linear process, its elements could be practically integrated with the Lean Six Sigma methodology. In the mapping above, both Lean Six Sigma and Design Thinking have a laser like focus on the Voice of the Customer (VOC). In Lean Six Sigma, we use techniques such as Kano Analysis or Quality Function Deployment to define our VOC's and CTQ's (Critical to Quality). Design Thinking, on the other hand, has a powerful technique of unearthing customers' or users' requirements called the Empathy Mapping. Using these approaches in an organized manner will make VOC determination more defined, more so, close to being granular.

The framework by which Design Thinking operates has the same simple construct as that of the PDCA cycle, and close semblance with the DMAIC methodology. Mapping the three models, here's how they intersect:

Lean Six Sigma	Design Thinking	PDCA Cycle
Define	Define and Understand	Plan
Measure	(Not much focus on Data Analysis)	
Analyze		
Improve	Ideate and Prototype	Do
Control	Test	Check
		Act

[The Crossroads: Lean Six Sigma and Design Thinking](#)

Lean Six Sigma, Design Thinking, and Agile can be regarded as complementary methodologies as they are compatible with one another in the areas of problem-solving and improvements. Fundamentally, each of these methodologies addresses different parts of the problem and lifecycle of a product or service.

When a problem exists but you cannot be certain of it, Design Thinking can help bring everyone together to follow a creative process to identify the problem as well as generate ideas that meet customer needs. Then, Lean Six Sigma through DMAIC approach can be employed to clarify the problem, determine the causes, and select the best solutions to rectify the problem to the existing process. Finally, once the best solution is chosen and involves any form of software development, Agile can be implemented by formulating user stories have the development team sprint to complete the development of those user stories to implement the solution.

Another way of integration Design Thinking tools can be used to determine and understand issues faced by customers to gain better customer behavioral insights and

become good input for the Agile development process and ensure the right customer issue is solved. When a product/service is developed through the Agile process, the Lean approach can help evaluate and improve it to spot and eliminate waste in the value stream for improved value. Not only that, Lean can also provide input for the Design Thinking process to better understand current “pain points” in existing services/products.

[Integration Of Lean Six Sigma, Design Thinking, And Agile & How They Complement Each Other](#)

Innovation and process management at Google

Innovation and Process Improvement at Google

■ Add bookmark

Tags: Alessandro Laureani Google Lean Six Sigma Internet Innovation

Innovation and process improvement go hand in hand, says Alessandro Laureani, Global Operations Processes and Systems Manager at what might be considered one of the most innovative companies on this planet...Google. There's a lot that needs to happen between the "Eureka" moment and the launch of a new service, he argues. In this Process Perspectives interview, Laureani discusses how the web giant is using Lean Six Sigma in their Adwords division and the challenges of improving processes...

- Six Sigma yellow and green belt certification
- To improve customer facing processes
- Many people think innovation and lean six sigma don't work together. I don't agree with that. We can streamline processes and eliminate waste. That gives us more time to look into real creativity. It also helps us in between the ideation and go-to-market.
- The challenge is to improve something that is continuously changing and also for highly creative people to learn the process skills.
- Something we do often is instead of immediately launching something that is global globally we launch it in a particular region or country first. If it is successful, we roll-out to the rest ([minimal viable segment](#) and agile)

- Inventing a new business process is considered innovation these days. I would challenge that being process-centric and being innovative are actually differences in that respect.
- My advice when working on a new process is to always keep the customer in mind and design processes that break down siloes inside the company.

[Podcast requires sign up \(free\)](#) (13 minutes)